

A general theory of the mechanics of petroleum reservoirs: geomechanics, compositional flow, reactions, and phase transformations

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June 23, 2026

Abstract

TBD

1 Material form of conservation of mass

Given a differential mass dm_ξ^α of component α in phase ξ , define the component partial density per current mixture volume by

$$\rho_\xi^\alpha(\mathbf{x}, t) = \frac{dm_\xi^\alpha}{dv}, \quad (1)$$

where $dv = dx_1 dx_2 dx_3$ is a current-configuration volume element of the mixture. The current configuration is a common mixture volume occupied by all components simultaneously. Each phase ξ motion is tracked from its own reference coordinates $\mathbf{X}_\xi$ and reference volume element

$$dV_\xi = (dX_1)_\xi (dX_2)_\xi (dX_3)_\xi. \quad (2)$$

The phase reference coordinates are related to the common current coordinates by a one-to-one, invertible, and continuously differentiable map on the region where that phase is present,

$$\mathbf{x} = \chi_\xi(\mathbf{X}_\xi, t), \quad (3)$$

with phase velocity $\mathbf{v}_\xi = \partial \chi_\xi / \partial t$ and Jacobian

$$J_\xi = \det \left(\frac{\partial \chi_\xi}{\partial \mathbf{X}_\xi} \right). \quad (4)$$

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Let $\rho_{\xi 0}^\alpha(\mathbf{X}_\xi)$ denote the initial (evaluated at $\mathbf{x}(t = 0) = \mathbf{X}_\xi$) component partial density, and let $\mathcal{C}_\xi^\alpha(\mathbf{X}_\xi, t)$ denote the mass of component α added to phase ξ per unit phase reference volume by any internal conversion mechanism. These mechanisms include both chemical reactions and pure phase transformations. Conservation of mass for component α in phase ξ over an arbitrary material region can then be written as

$$\int_{\mathcal{B}_{\xi 0}} (\rho_{\xi 0}^\alpha + \mathcal{C}_\xi^\alpha) dV_\xi = \int_{\mathcal{B}} \rho_\xi^\alpha dv = \int_{\mathcal{B}_{\xi 0}} \rho_\xi^\alpha J_\xi dV_\xi. \quad (5)$$

Here $\mathcal{B}$ is the common current mixture region corresponding, through the phase- ξ map, to the phase reference region $\mathcal{B}_{\xi 0}$. Since (5) must hold for arbitrary $\mathcal{B}_{\xi 0}$,

$$\rho_{\xi 0}^\alpha + \mathcal{C}_\xi^\alpha = \rho_\xi^\alpha J_\xi. \quad (6)$$

The material time derivative of a quantity $\square_\xi$ following the velocity of phase ξ is

$$\frac{D_\xi \square_\xi}{Dt} = \frac{\partial \square_\xi}{\partial t} + \mathbf{v}_\xi \cdot \nabla_{\mathbf{x}} \square_\xi, \quad (7)$$

and we use the shorthand $\dot{\square}_\xi = D_\xi \square_\xi / Dt$ for the material time derivative following phase ξ . Taking the material time derivative of (6) following phase ξ and using $\dot{J}_\xi = J_\xi \nabla_{\mathbf{x}} \cdot \mathbf{v}_\xi$ gives

$$\frac{D_\xi \rho_\xi^\alpha}{Dt} + \rho_\xi^\alpha \nabla_{\mathbf{x}} \cdot \mathbf{v}_\xi = \frac{\dot{\mathcal{C}}_\xi^\alpha}{J_\xi}. \quad (8)$$

We define the current production rate per unit current mixture volume by

$$\dot{c}_\xi^\alpha(\mathbf{x}, t) = \frac{\dot{\mathcal{C}}_\xi^\alpha}{J_\xi}. \quad (9)$$

Expanding (8) gives the spatial component mass balance of the ξ phase

$$\frac{\partial \rho_\xi^\alpha}{\partial t} + \nabla_{\mathbf{x}} \cdot (\rho_\xi^\alpha \mathbf{v}_\xi) = \dot{c}_\xi^\alpha. \quad (10)$$

Now introduce the phase partial density $\rho_\xi = \sum_\alpha \rho_\xi^\alpha$ and the mass fraction of component α in phase ξ ,

$$\eta_\xi^\alpha = \frac{\rho_\xi^\alpha}{\rho_\xi}, \quad \sum_\alpha \eta_\xi^\alpha = 1. \quad (11)$$

The phase volume fraction is defined by $\phi_\xi = dv_\xi / dv$, and the intrinsic density by $\bar{\rho}_\xi = dm_\xi / dv_\xi$, so that

$$\rho_\xi = \phi_\xi \bar{\rho}_\xi, \quad \rho_\xi^\alpha = \phi_\xi \bar{\rho}_\xi \eta_\xi^\alpha. \quad (12)$$

Substituting this representation into (10) gives

$$\frac{\partial}{\partial t} (\phi_\xi \bar{\rho}_\xi \eta_\xi^\alpha) + \nabla_{\mathbf{x}} \cdot (\phi_\xi \bar{\rho}_\xi \eta_\xi^\alpha \mathbf{v}_\xi) = \dot{c}_\xi^\alpha. \quad (13)$$

All internal conversion mechanisms are parameterized by current progress rates $\dot{r}_m$ and phase/component stoichiometric coefficients $\nu_{\xi m}^\alpha$:

$$\dot{c}_\xi^\alpha = \sum_m \nu_{\xi m}^\alpha \dot{r}_m. \quad (14)$$

The index m includes chemical reactions and pure phase transformations. A pure phase transformation of component α from phase ξ to phase ζ is the special one-to-one stoichiometry $\nu_{\xi m}^\alpha = -1$, $\nu_{\zeta m}^\alpha = +1$, with all other entries zero. A chemical reaction may change both component identity and phase label. If each mechanism conserves total mass, $\sum_\xi \sum_\alpha \nu_{\xi m}^\alpha = 0$, then the overall mixture mass is conserved automatically. Component conservation is recovered for those pure phase-transformation mechanisms whose stoichiometry has zero sum over phases for the fixed component.

2 Variational Derivation

We derive the governing equations from Hamilton's extended principle, following Bedford & Drumheller [2], adapted to a compositional mixture system containing phases (solid, aqueous, oliec, and gas phases) and arbitrary miscible components α within each phase.

Each phase has volume fraction ϕ_ξ , intrinsic density $\bar{\rho}_\xi$, bulk density $\rho_\xi = \phi_\xi \bar{\rho}_\xi$, velocity $\mathbf{v}_\xi$, deformation gradient $\mathbf{F}_\xi$, and Jacobian $J_\xi = \det(\mathbf{F}_\xi)$.

2.1 Variational principle and its terms

The extended Hamilton's principle is

$$\int_{t_1}^{t_2} \left[\delta(T - U) + \delta W + \sum_k \delta C_k \right] dt = 0,$$

with kinetic energy, internal energy, virtual work, and four constraints:

$$T = \sum_\xi \int_B \frac{1}{2} \rho_\xi \mathbf{v}_\xi \cdot \mathbf{v}_\xi dv, \quad (15)$$

$$U = \sum_\xi \int_B \rho_\xi \psi_\xi(\mathbf{F}_\xi, \bar{\rho}_\xi, \eta_\xi^\alpha) dv + \int_B \phi \gamma(S_a, S_g) dv, \quad (16)$$

$$\delta W = \sum_\xi \int_B (\rho_\xi \mathbf{g} + \mathbf{b}_\xi) \cdot \delta \mathbf{x}_\xi dv - \sum_\xi \sum_\alpha \int_B L_\xi^\alpha \frac{\delta \mathcal{C}_\xi^\alpha}{J_\xi} dv - \sum_m \int_B \mathcal{A}_m \delta \dot{r}_m dv, \quad (17)$$

$$C_1 = \int_B \lambda \left(\sum_\xi \phi_\xi - 1 \right) dv, \quad (18)$$

$$C_2 = \sum_\xi \sum_\alpha \int_{B_{\xi 0}} \tau_\xi^\alpha \left(J_\xi - \frac{\phi_{\xi 0} \eta_{\xi 0}^\alpha \bar{\rho}_{\xi 0} + \mathcal{C}_\xi^\alpha}{\phi_\xi \eta_\xi^\alpha \bar{\rho}_\xi} \right) dV_\xi, \quad (19)$$

$$C_3 = \sum_\xi \int_B \pi_\xi \left(\sum_\alpha \eta_\xi^\alpha - 1 \right) dv, \quad (20)$$

$$C_4 = \sum_\xi \sum_\alpha \int_B \mu_\xi^\alpha \left(\frac{\dot{\mathcal{C}}_\xi^\alpha}{J_\xi} - \sum_m \nu_{\xi m}^\alpha \dot{r}_m \right) dv. \quad (21)$$

The coefficients L_ξ^α are component conversion-work potentials conjugate to adding component mass to phase ξ . The affinities $\mathcal{A}_m$ are introduced as generalized mechanism forces in the virtual-work one-form, conjugate to the current progress-rate variations $\delta \dot{r}_m$. The kinematic constraint C_4 ties those progress rates to the phase-attached accumulated sources through $\dot{c}_\xi^\alpha = \dot{\mathcal{C}}_\xi^\alpha / J_\xi$. This

distinction is essential: $\dot{c}_\xi^\alpha$ is not a freely varied Eulerian scalar, because variation of the phase map also varies J_ξ . Chemical reactions and pure phase transformations are both represented by the unified stoichiometric parameterization (14). We treat $\mathbf{x}_\xi$, η_ξ^α , ϕ_ξ , $\bar{\rho}_\xi$, $\mathcal{C}_\xi^\alpha$, $\dot{r}_m$, μ_ξ^α , and $\mathcal{A}_m$ as variational variables. The conversion-work potentials L_ξ^α are constitutive inputs, with additional closure required for kinetically limited mechanisms.

2.2 Extending Bedford and Drumheller identity 1.28

The accumulated source $\mathcal{C}_\xi^\alpha$ is attached to the phase- ξ reference coordinates, while its current production rate is $\dot{c}_\xi^\alpha = \dot{\mathcal{C}}_\xi^\alpha / J_\xi$. Thus variations of source terms must be pushed forward with the same phase map as the corresponding component density. For any phase field Γ_ξ , the Bedford–Drumheller identity extends componentwise as

$$\delta \int_B \rho_\xi^\alpha \Gamma_\xi \, dv = \int_B \left(\rho_\xi^\alpha \delta \Gamma_\xi + \frac{\delta \mathcal{C}_\xi^\alpha}{J_\xi} \Gamma_\xi \right) dv. \quad (22)$$

summing both sides of (22) over α gives

$$\delta \int_B \rho_\xi \Gamma_\xi \, dv = \int_B \left(\rho_\xi \delta \Gamma_\xi + \sum_\alpha \frac{\delta \mathcal{C}_\xi^\alpha}{J_\xi} \Gamma_\xi \right) dv \quad (23)$$

where we've used $\rho_\xi = \sum_\alpha \rho_\xi^\alpha$.

2.3 Variation of the kinetic energy

The variation of kinetic energy using (23) is then

$$\delta T = \sum_\xi \int_B \rho_\xi \mathbf{v}_\xi \cdot \delta \mathbf{v}_\xi + \frac{1}{2} \mathbf{v}_\xi \cdot \mathbf{v}_\xi \sum_\alpha \frac{\delta \mathcal{C}_\xi^\alpha}{J_\xi} dv. \quad (24)$$

Since $\delta \mathbf{v}_\xi = d(\delta \mathbf{x}_\xi)/dt$ and $\delta \mathbf{x}_\xi|_{t_1} = \delta \mathbf{x}_\xi|_{t_2} = \mathbf{0}$, integration by parts in time gives

$$\int_{t_1}^{t_2} \delta T \, dt = \int_{t_1}^{t_2} \sum_\xi \int_B \left(-\rho_\xi \mathbf{a}_\xi - \sum_\alpha \dot{c}_\xi^\alpha \mathbf{v}_\xi \right) \cdot \delta \mathbf{x}_\xi + \frac{1}{2} \mathbf{v}_\xi \cdot \mathbf{v}_\xi \sum_\alpha \frac{\delta \mathcal{C}_\xi^\alpha}{J_\xi} dv \, dt \quad (25)$$

2.4 Variation of the phase free energies

For every phase, solid or mobile, let ψ_ξ denote the specific Helmholtz free energy. The corresponding contribution to the total free energy is written in the same current density-weighted form,

$$\begin{aligned} U_\xi &= \sum_\xi \int_B \rho_\xi \psi_\xi(\mathbf{F}_\xi, \bar{\rho}_\xi, \eta_\xi^1, \eta_\xi^2, \dots, \eta_\xi^N) \, dv \\ &= \sum_\xi \int_B \rho_\xi \psi_\xi(\mathbf{F}_\xi, \bar{\rho}_\xi, \eta_\xi^\beta) \, dv \end{aligned} \quad (26)$$

Here η_ξ^β denotes the full list of component mass fractions $\{\eta_\xi^1, \dots, \eta_\xi^N\}$; the index β is not summed in this argument list. The constitutive dependence on $\mathbf{F}_\xi$ may be trivial for mobile phases, but keeping it in the notation allows one calculation for all phases. The local free-energy variation is

$$\delta \psi_\xi = \left. \frac{\partial \psi_\xi}{\partial \mathbf{F}_\xi} \right|_{\bar{\rho}_\xi, \eta_\xi^\beta} : \delta \mathbf{F}_\xi + \left. \frac{\partial \psi_\xi}{\partial \bar{\rho}_\xi} \right|_{\mathbf{F}_\xi, \eta_\xi^\beta} \delta \bar{\rho}_\xi + \sum_\alpha \left. \frac{\partial \psi_\xi}{\partial \eta_\xi^\alpha} \right|_{\mathbf{F}_\xi, \bar{\rho}_\xi, \eta_\xi^{\beta \neq \alpha}} \delta \eta_\xi^\alpha. \quad (27)$$

Using (23), integrating the $\delta \mathbf{F}_\xi$ term by parts, and collecting independent variations gives

$$\begin{aligned} \delta U_\xi = \sum_\xi \int_B \left[-\nabla_{\mathbf{x}} \cdot \boldsymbol{\sigma}'_\xi \cdot \delta \mathbf{x}_\xi + \rho_\xi \frac{\partial \psi_\xi}{\partial \bar{\rho}_\xi} \Big|_{\mathbf{F}_\xi, \eta_\xi^\beta} \delta \bar{\rho}_\xi \right. \\ \left. + \sum_\alpha \rho_\xi \frac{\partial \psi_\xi}{\partial \eta_\xi^\alpha} \Big|_{\mathbf{F}_\xi, \bar{\rho}_\xi, \eta_\xi^{\beta \neq \alpha}} \delta \eta_\xi^\alpha + \sum_\alpha \psi_\xi \frac{\delta \mathcal{C}_\xi^\alpha}{J_\xi} \right] dv + \text{boundary terms.} \end{aligned} \quad (28)$$

The corresponding Cauchy stress is

$$\boldsymbol{\sigma}'_\xi \equiv \rho_\xi \frac{\partial \psi_\xi}{\partial \mathbf{F}_\xi} \Big|_{\bar{\rho}_\xi, \eta_\xi^\beta} \mathbf{F}_\xi^T. \quad (29)$$

For mobile phases this term vanishes when ψ_ξ has no dependence on $\mathbf{F}_\xi$.

2.5 Variation of the interfacial energies

With three mobile phases (aqueous, oleic, and gas), let

$$\phi = \phi_a + \phi_o + \phi_g, \quad S_a = \frac{\phi_a}{\phi}, \quad S_g = \frac{\phi_g}{\phi}, \quad (30)$$

so that $S_o = 1 - S_a - S_g = \phi_o/\phi$. The saturations are used here only as ratios of the primitive volume-fraction fields; the variations are still taken with respect to ϕ_a , ϕ_o , and ϕ_g . We specialize the interfacial energy density to

$$U_\gamma = \int_B \phi \gamma(S_a, S_g) \, dv. \quad (31)$$

Following the saturation-based capillary-pressure construction used for the two-mobile-phase case, but now taking oil as the reference mobile phase, define

$$p_{c,oa} \equiv -\frac{\partial \gamma}{\partial S_a}, \quad p_{c,og} \equiv -\frac{\partial \gamma}{\partial S_g}. \quad (32)$$

Here $p_{c,oa} = \bar{p}_o - \bar{p}_a$ is the oil–aqueous capillary pressure and $p_{c,og} = \bar{p}_o - \bar{p}_g$ is the additional oil–gas capillary pressure. Using $\delta \phi = \delta \phi_a + \delta \phi_o + \delta \phi_g$ and

$$\delta S_a = \frac{\delta \phi_a - S_a \delta \phi}{\phi}, \quad \delta S_g = \frac{\delta \phi_g - S_g \delta \phi}{\phi}, \quad (33)$$

the algebraic part of the interfacial variation can be written as

$$\begin{aligned} \delta(\phi \gamma) &= \gamma \delta \phi + \frac{\partial \gamma}{\partial S_a} (\delta \phi_a - S_a \delta \phi) + \frac{\partial \gamma}{\partial S_g} (\delta \phi_g - S_g \delta \phi) \\ &= \gamma_a \delta \phi_a + \gamma_o \delta \phi_o + \gamma_g \delta \phi_g, \end{aligned} \quad (34)$$

where

$$\begin{aligned} \gamma_o &= \gamma + S_a p_{c,oa} + S_g p_{c,og}, \\ \gamma_a &= \gamma_o - p_{c,oa}, \\ \gamma_g &= \gamma_o - p_{c,og}. \end{aligned} \quad (35)$$

Consequently the capillary pressures are the oil-referenced differences of the primitive interfacial potentials,

$$p_{c,oa} \equiv \bar{p}_o - \bar{p}_a \equiv \gamma_o - \gamma_a, \quad p_{c,og} \equiv \bar{p}_o - \bar{p}_g \equiv \gamma_o - \gamma_g. \quad (36)$$

The spatial fields must be varied using the inter-species transport identity. For the phase displacement $\delta \mathbf{x}_\xi$, the volume-fraction variation contains the transported part

$$\delta_x \phi_\xi = -\nabla_{\mathbf{x}} \cdot (\phi_\xi \delta \mathbf{x}_\xi), \quad (37)$$

in addition to the independent coefficient variation used in the algebraic Euler–Lagrange equations. Consequently,

$$\begin{aligned} \delta U_\gamma = & \int_{\mathcal{B}} (\gamma_a \delta \phi_a + \gamma_o \delta \phi_o + \gamma_g \delta \phi_g) \, dv \\ & + \int_{\mathcal{B}} [\phi_a \nabla_{\mathbf{x}} \gamma_a \cdot \delta \mathbf{x}_a + \phi_o \nabla_{\mathbf{x}} \gamma_o \cdot \delta \mathbf{x}_o + \phi_g \nabla_{\mathbf{x}} \gamma_g \cdot \delta \mathbf{x}_g] \, dv + \text{boundary terms}. \end{aligned} \quad (38)$$

This form keeps the gradient terms generated by the spatial dependence of the volume fractions. They cancel against the corresponding multiplier gradients in the phase momentum balances after the volume-fraction Euler–Lagrange equations are used.

2.6 Variation of constraint C_1 : volume-fraction

The constraint $C_1 = \int_{\mathcal{B}} \lambda (\sum_\xi \phi_\xi - 1) \, dv$ is written in the current configuration, so the natural variation is Eulerian, i.e. taken at fixed spatial point $\mathbf{x}$. Since each ϕ_ξ is a constituent-attached spatial field, we use the Bedford–Drumheller identity

$$\delta \phi_\xi|_{\mathbf{x}} = \delta \phi_\xi - \nabla_{\mathbf{x}} \phi_\xi \cdot \delta \mathbf{x}_\xi, \quad (39)$$

to express the variation in terms of the spatial variation $\delta \phi_\xi$ and the motion variation $\delta \mathbf{x}_\xi$. Therefore

$$\delta C_1 = \int_{\mathcal{B}} \delta \lambda \left(\sum_\xi \phi_\xi - 1 \right) \, dv + \sum_\xi \int_{\mathcal{B}} \lambda (\delta \phi_\xi - \nabla_{\mathbf{x}} \phi_\xi \cdot \delta \mathbf{x}_\xi) \, dv. \quad (40)$$

The first term enforces the saturation constraint $\sum_\xi \phi_\xi = 1$. The remaining terms contribute λ to each $\delta \phi_\xi$ equation and $-\lambda \nabla_{\mathbf{x}} \phi_\xi$ to each momentum variation $\delta \mathbf{x}_\xi$.

2.7 Variation of constraint C_2 : component material mass balance

The component material mass balance constraint is

$$C_2 = \sum_\xi \sum_\alpha \int_{\mathcal{B}_{\xi 0}} \tau_\xi^\alpha \left(J_\xi - \frac{\phi_{\xi 0} \bar{\rho}_{\xi 0} \eta_{\xi 0}^\alpha + \mathcal{C}_\xi^\alpha}{\phi_\xi \bar{\rho}_\xi \eta_\xi^\alpha} \right) dV_\xi. \quad (41)$$

Hence

$$\begin{aligned}
\delta C_2 = & \sum_{\xi} \sum_{\alpha} \int_{\mathcal{B}_{\xi 0}} \tau_{\xi}^{\alpha} \delta J_{\xi} dV_{\xi} \\
& + \sum_{\xi} \sum_{\alpha} \int_{\mathcal{B}_{\xi 0}} \tau_{\xi}^{\alpha} \frac{\phi_{\xi 0} \bar{\rho}_{\xi 0} \eta_{\xi 0}^{\alpha} + \mathcal{C}_{\xi}^{\alpha}}{\phi_{\xi}^2 \bar{\rho}_{\xi} \eta_{\xi}^{\alpha}} \delta \phi_{\xi} dV_{\xi} \\
& + \sum_{\xi} \sum_{\alpha} \int_{\mathcal{B}_{\xi 0}} \tau_{\xi}^{\alpha} \frac{\phi_{\xi 0} \bar{\rho}_{\xi 0} \eta_{\xi 0}^{\alpha} + \mathcal{C}_{\xi}^{\alpha}}{\phi_{\xi} \bar{\rho}_{\xi}^2 \eta_{\xi}^{\alpha}} \delta \bar{\rho}_{\xi} dV_{\xi} \\
& + \sum_{\xi} \sum_{\alpha} \int_{\mathcal{B}_{\xi 0}} \tau_{\xi}^{\alpha} \frac{\phi_{\xi 0} \bar{\rho}_{\xi 0} \eta_{\xi 0}^{\alpha} + \mathcal{C}_{\xi}^{\alpha}}{\phi_{\xi} \bar{\rho}_{\xi} (\eta_{\xi}^{\alpha})^2} \delta \eta_{\xi}^{\alpha} dV_{\xi} \\
& - \sum_{\xi} \sum_{\alpha} \int_{\mathcal{B}_{\xi 0}} \frac{\tau_{\xi}^{\alpha}}{\phi_{\xi} \bar{\rho}_{\xi} \eta_{\xi}^{\alpha}} \delta \mathcal{C}_{\xi}^{\alpha} dV_{\xi}.
\end{aligned} \tag{42}$$

Using $dv = J_{\xi} dV_{\xi}$ and the constraint $\phi_{\xi 0} \bar{\rho}_{\xi 0} \eta_{\xi 0}^{\alpha} + \mathcal{C}_{\xi}^{\alpha} = J_{\xi} \phi_{\xi} \bar{\rho}_{\xi} \eta_{\xi}^{\alpha}$, the individual contributions push forward as follows.

δJ_{ξ} terms (from $\delta \mathbf{x}_{\xi}$):

$$\sum_{\xi} \sum_{\alpha} \int_{\mathcal{B}_{\xi 0}} \tau_{\xi}^{\alpha} \delta J_{\xi} dV_{\xi} = - \sum_{\xi} \sum_{\alpha} \int_{\mathcal{B}} \nabla_{\mathbf{x}} \tau_{\xi}^{\alpha} \cdot \delta \mathbf{x}_{\xi} dv + \text{b.t.} \tag{43}$$

Spatial field terms:

$$\delta C_2 \supset \sum_{\xi} \sum_{\alpha} \int_{\mathcal{B}} \left[\frac{\tau_{\xi}^{\alpha}}{\phi_{\xi}} \delta \phi_{\xi} + \frac{\tau_{\xi}^{\alpha}}{\bar{\rho}_{\xi}} \delta \bar{\rho}_{\xi} + \frac{\tau_{\xi}^{\alpha}}{\eta_{\xi}^{\alpha}} \delta \eta_{\xi}^{\alpha} - \frac{\tau_{\xi}^{\alpha}}{\phi_{\xi} \bar{\rho}_{\xi} \eta_{\xi}^{\alpha}} \frac{\delta \mathcal{C}_{\xi}^{\alpha}}{J_{\xi}} \right] dv. \tag{44}$$

2.8 Variation of constraint C_3 : mass partition within phase

$$\begin{aligned}
\delta C_3 = & \sum_{\xi} \int_{\mathcal{B}} \delta \pi_{\xi} \left(\sum_{\alpha} \eta_{\xi}^{\alpha} - 1 \right) dv + \sum_{\xi} \sum_{\alpha} \int_{\mathcal{B}} \pi_{\xi} \left(\delta \eta_{\xi}^{\alpha} - \nabla_{\mathbf{x}} \eta_{\xi}^{\alpha} \cdot \delta \mathbf{x}_{\xi} \right) dv, \\
= & \sum_{\xi} \int_{\mathcal{B}} \delta \pi_{\xi} \left(\sum_{\alpha} \eta_{\xi}^{\alpha} - 1 \right) dv + \sum_{\xi} \sum_{\alpha} \int_{\mathcal{B}} \pi_{\xi} \delta \eta_{\xi}^{\alpha} dv.
\end{aligned} \tag{45}$$

The first line vanishes on the constraint surface $\sum_{\alpha} \eta_{\xi}^{\alpha} = 1$ and recovers it as an Euler–Lagrange equation. The second line contributes π_{ξ} to each $\delta \eta_{\xi}^{\alpha}$ Euler–Lagrange equation.

2.9 Stoichiometric conversion variations

Conversion sources are restricted by the current-rate kinematic constraint (21). Define the component conversion potential coefficient The variation of $\dot{c}_{\xi}^{\alpha} = \dot{\mathcal{C}}_{\xi}^{\alpha} / J_{\xi}$ must be taken before using the stoichiometric projection, because the denominator depends on the phase map. Repeating the

transport calculation used for phase-attached rates gives

$$\begin{aligned}
\int_{t_1}^{t_2} \delta C_4 \, dt &= \int_{t_1}^{t_2} \int_{\mathcal{B}} \sum_{\xi} \sum_{\alpha} \delta \mu_{\xi}^{\alpha} \left(\dot{c}_{\xi}^{\alpha} - \sum_m \nu_{\xi m}^{\alpha} \dot{r}_m \right) dv \, dt \\
&\quad + \int_{t_1}^{t_2} \int_{\mathcal{B}} \sum_{\xi} \sum_{\alpha} \left[-\frac{D_{\xi} \mu_{\xi}^{\alpha}}{Dt} \frac{\delta \mathcal{C}_{\xi}^{\alpha}}{J_{\xi}} + \dot{c}_{\xi}^{\alpha} \nabla_{\mathbf{x}} \mu_{\xi}^{\alpha} \cdot \delta \mathbf{x}_{\xi} \right] dv \, dt \\
&\quad - \int_{t_1}^{t_2} \int_{\mathcal{B}} \sum_m \left(\sum_{\xi} \sum_{\alpha} \mu_{\xi}^{\alpha} \nu_{\xi m}^{\alpha} \right) \delta \dot{r}_m \, dv \, dt + \text{b.t.}
\end{aligned} \tag{46}$$

The affinity is not introduced by allowing $\dot{c}_{\xi}^{\alpha}$ to vary freely. The component-level conversion work has already entered through (17); $\mathcal{A}_m$ is the generalized mechanism force conjugate to the progress-rate variation $\delta \dot{r}_m$. Its stoichiometric identification is obtained below as the Euler–Lagrange equation associated with $\dot{r}_m$.

Thus μ_{ξ}^{α} is a kinematic multiplier enforcing the phase-attached push-forward and stoichiometry, while $\mathcal{A}_m$ is the thermodynamic force conjugate to mechanism progress. The term $\dot{c}_{\xi}^{\alpha} \nabla \mu_{\xi}^{\alpha}$ is the part of the mechanical coupling that would be lost if $\delta \dot{c}_{\xi}^{\alpha}$ were treated as a free Eulerian variation.

2.10 Collected variational principle

Evaluating all variations and collecting by independent variations $\delta \mathbf{x}_{\xi}$, $\delta \phi_{\xi}$, $\delta \bar{\rho}_{\xi}$, $\delta \eta_{\xi}^{\alpha}$, and $\delta \dot{r}_m$ gives the following reduced collection. In the displacement bracket, the raw multiplier combination $-\lambda \nabla_{\mathbf{x}} \phi_{\xi} - \nabla_{\mathbf{x}} \sum_{\alpha} \tau_{\xi}^{\alpha}$ has been combined into $\phi_{\xi} \nabla_{\mathbf{x}} \lambda$ using the corresponding volume-fraction coefficient; the mobile-phase capillary terms are displayed separately.

$$\begin{aligned}
0 &= \sum_{\xi} \int_{\mathcal{B}} \left[-\rho_{\xi} \mathbf{a}_{\xi} + \nabla_{\mathbf{x}} \cdot \boldsymbol{\sigma}'_{\xi} + \rho_{\xi} \mathbf{g} + \mathbf{b}_{\xi} + \phi_{\xi} \nabla_{\mathbf{x}} \lambda + \sum_{\alpha} \dot{c}_{\xi}^{\alpha} \left(\nabla_{\mathbf{x}} \mu_{\xi}^{\alpha} - \mathbf{v}_{\xi} \right) \right] \cdot \delta \mathbf{x}_{\xi} \, dv \\
&\quad - \int_{\mathcal{B}} [\phi_a \nabla_{\mathbf{x}} \gamma_a \cdot \delta \mathbf{x}_a + \phi_o \nabla_{\mathbf{x}} \gamma_o \cdot \delta \mathbf{x}_o + \phi_g \nabla_{\mathbf{x}} \gamma_g \cdot \delta \mathbf{x}_g] \, dv \\
&\quad + \sum_{\xi} \int_{\mathcal{B}} \left[\lambda + \sum_{\alpha} \frac{\tau_{\xi}^{\alpha}}{\phi_{\xi}} \right] \delta \phi_{\xi} \, dv \\
&\quad - \int_{\mathcal{B}} (\gamma_a \delta \phi_a + \gamma_o \delta \phi_o + \gamma_g \delta \phi_g) \, dv \\
&\quad + \sum_{\xi} \int_{\mathcal{B}} \left[-\rho_{\xi} \left. \frac{\partial \psi_{\xi}}{\partial \bar{\rho}_{\xi}} \right|_{\mathbf{F}_{\xi}, \eta_{\xi}^{\beta}} + \sum_{\alpha} \frac{\tau_{\xi}^{\alpha}}{\bar{\rho}_{\xi}} \right] \delta \bar{\rho}_{\xi} \, dv \\
&\quad + \sum_{\xi} \sum_{\alpha} \int_{\mathcal{B}} \left[\frac{\tau_{\xi}^{\alpha}}{\eta_{\xi}^{\alpha}} + \pi_{\xi} - \rho_{\xi} \left. \frac{\partial \psi_{\xi}}{\partial \eta_{\xi}^{\alpha}} \right|_{\mathbf{F}_{\xi}, \bar{\rho}_{\xi}, \eta_{\xi}^{\beta \neq \alpha}} \right] \delta \eta_{\xi}^{\alpha} \, dv \\
&\quad + \sum_{\xi} \sum_{\alpha} \int_{\mathcal{B}} \left[\frac{1}{2} \mathbf{v}_{\xi} \cdot \mathbf{v}_{\xi} - \psi_{\xi} - L_{\xi}^{\alpha} - \frac{D_{\xi} \mu_{\xi}^{\alpha}}{Dt} - \frac{\tau_{\xi}^{\alpha}}{\phi_{\xi} \bar{\rho}_{\xi} \eta_{\xi}^{\alpha}} \right] \frac{\delta \mathcal{C}_{\xi}^{\alpha}}{J_{\xi}} \, dv \\
&\quad - \sum_m \int_{\mathcal{B}} \left(\mathcal{A}_m + \sum_{\xi} \sum_{\alpha} \mu_{\xi}^{\alpha} \nu_{\xi m}^{\alpha} \right) \delta \dot{r}_m \, dv + \text{boundary terms.}
\end{aligned} \tag{47}$$

The multiplier variations are omitted here because they return the constraint equations (18)–(21). Since all variations are considered arbitrary, each bracketed integrand must vanish pointwise, yielding the Euler–Lagrange equations.

2.11 Euler–Lagrange equations

The Euler–Lagrange equations generated by arbitrary $\delta\phi_\xi$ give, for phases without an explicit capillary contribution,

$$\phi_\xi \lambda = - \sum_\alpha \tau_\xi^\alpha. \quad (48)$$

For the mobile phases the corresponding equations contain the primitive capillary potentials discussed below.

The displacement equation obtained from the uniform part of the collected principle is

$$\rho_\xi \mathbf{a}_\xi = \nabla_{\mathbf{x}} \cdot \boldsymbol{\sigma}'_\xi + \rho_\xi \mathbf{g} + \mathbf{b}_\xi + \phi_\xi \nabla_{\mathbf{x}} \lambda + \sum_\alpha \dot{c}_\xi^\alpha \left(\nabla_{\mathbf{x}} \mu_\xi^\alpha - \mathbf{v}_\xi \right), \quad (49)$$

with the mobile-phase capillary-force additions displayed separately below.

The Euler–Lagrange equation associated with arbitrary $\delta\bar{\rho}_\xi$ gives

$$\sum_\alpha \frac{\tau_\xi^\alpha}{\phi_\xi} = \bar{\rho}_\xi^2 \left. \frac{\partial \psi_\xi}{\partial \bar{\rho}_\xi} \right|_{\mathbf{F}_\xi, \eta_\xi^\beta} \equiv \bar{p}_\xi. \quad (50)$$

These two Euler–Lagrange relations also recover the capillary-pressure construction for the mobile phases. Let $l \in \{a, o, g\}$ denote a mobile phase. The coefficient of $\delta\phi_l$ in (47) includes the primitive interfacial potential γ_l , so the mobile-phase volume-fraction relation is

$$\lambda + \sum_\alpha \frac{\tau_l^\alpha}{\phi_l} - \gamma_l = 0. \quad (51)$$

Using (50) gives

$$\lambda + \bar{p}_l - \gamma_l = 0, \quad \text{or} \quad \bar{p}_l = \gamma_l - \lambda. \quad (52)$$

Consequently pressure differences between mobile phases are exactly the differences of the primitive interfacial potentials,

$$\bar{p}_l - \bar{p}_k = \gamma_l - \gamma_k. \quad (53)$$

For the oil-referenced definitions above, this gives

$$\begin{aligned} \bar{p}_o - \bar{p}_a &= \gamma_o - \gamma_a \\ &= p_{c,oa}, \\ \bar{p}_o - \bar{p}_g &= \gamma_o - \gamma_g \\ &= p_{c,og}. \end{aligned} \quad (54)$$

The same relations identify the equivalent pore pressure associated with the volume constraint. Define

$$\bar{p}_E \equiv -\lambda. \quad (55)$$

Multiplying $\bar{p}_l = \gamma_l + \bar{p}_E$ by the mobile saturations and summing over mobile phases gives

$$\sum_l S_l \bar{p}_l = \bar{p}_E + \sum_l S_l \gamma_l. \quad (56)$$

The primitive potentials satisfy

$$\sum_l S_l \gamma_l = S_a(\gamma_o - p_{c,oa}) + S_o \gamma_o + S_g(\gamma_o - p_{c,og}) = \gamma. \quad (57)$$

Therefore

$$\bar{p}_E = \sum_l S_l \bar{p}_l - \gamma. \quad (58)$$

The Euler–Lagrange equations associated with arbitrary $\delta\eta_\xi^\alpha$ give

$$0 = \frac{\tau_\xi^\alpha}{\eta_\xi^\alpha} + \pi_\xi - \rho_\xi \left. \frac{\partial \psi_\xi}{\partial \eta_\xi^\alpha} \right|_{\mathbf{F}_\xi, \bar{\rho}_\xi, \eta_\xi^{\beta \neq \alpha}}, \quad \alpha = 1, \dots, N, \quad (59)$$

while variation with respect to π_ξ returns the composition normalization constraint

$$\sum_{\alpha=1}^N \eta_\xi^\alpha = 1. \quad (60)$$

Thus π_ξ is the phase-level multiplier that enforces the redundant composition coordinate. Equation (59) may be read componentwise as

$$\rho_\xi \left. \frac{\partial \psi_\xi}{\partial \eta_\xi^\alpha} \right|_{\mathbf{F}_\xi, \bar{\rho}_\xi, \eta_\xi^{\beta \neq \alpha}} - \frac{\tau_\xi^\alpha}{\eta_\xi^\alpha} = \pi_\xi, \quad (61)$$

$$\rho_\xi \left(\left. \frac{\partial \psi_\xi}{\partial \eta_\xi^\alpha} \right|_{\mathbf{F}_\xi, \bar{\rho}_\xi, \eta_\xi^{\kappa \neq \alpha}} - \left. \frac{\partial \psi_\xi}{\partial \eta_\xi^\beta} \right|_{\mathbf{F}_\xi, \bar{\rho}_\xi, \eta_\xi^{\kappa \neq \beta}} \right) = \frac{\tau_\xi^\alpha}{\eta_\xi^\alpha} - \frac{\tau_\xi^\beta}{\eta_\xi^\beta}. \quad (62)$$

Here κ denotes the collection of composition components held fixed in the partial derivative; it is not summed.

Choosing $\alpha = 1$ and comparing against $j = 2, 3, \dots, N$ gives a system of $N - 1$ equations,

$$\rho_\xi \left(\left. \frac{\partial \psi_\xi}{\partial \eta_\xi^1} \right|_{\mathbf{F}_\xi, \bar{\rho}_\xi, \eta_\xi^{\kappa \neq 1}} - \left. \frac{\partial \psi_\xi}{\partial \eta_\xi^j} \right|_{\mathbf{F}_\xi, \bar{\rho}_\xi, \eta_\xi^{\kappa \neq j}} \right) = \frac{\tau_\xi^1}{\eta_\xi^1} - \frac{\tau_\xi^j}{\eta_\xi^j}, \quad j = 2, \dots, N. \quad (63)$$

The equations associated with the conversion variables are

$$\frac{D_\xi \mu_\xi^\alpha}{Dt} = \frac{1}{2} \mathbf{v}_\xi \cdot \mathbf{v}_\xi - \psi_\xi - L_\xi^\alpha - \frac{\tau_\xi^\alpha}{\phi_\xi \bar{\rho}_\xi \eta_\xi^\alpha}, \quad (64)$$

$$\mathcal{A}_m = - \sum_\xi \sum_\alpha \mu_\xi^\alpha \nu_{\xi m}^\alpha, \quad m = 1, \dots, N_m. \quad (65)$$

Equation (65) is not an additional closure relation for the state variables. It defines the mechanism affinity as the stoichiometric projection of the phase-attached conversion potentials, but only after the kinematic push-forward $\dot{c}_\xi^\alpha = \dot{C}_\xi^\alpha / J_\xi$ has been varied. The closure enters only when one prescribes either an equilibrium condition $\mathcal{A}_m = 0$ or a kinetic relation $\dot{r}_m = \mathcal{K}_m(\mathcal{A}_m, \text{state})$. Thus, for kinetically limited mechanisms such as slow fluid–solid reactions, (65) supplies the driving force rather than an algebraic equilibrium equation. For an instantaneous phase transformation, $\dot{r}_m$ may be a measure in time. In all cases, the phase momentum equations retain the inertial production term and the $\nabla \mu_\xi^\alpha$ force generated by adding or removing mass in a phase moving with velocity $\mathbf{v}_\xi$. For a single-component solid, $\eta_{\xi_s}^\alpha = 1$.

At this variational stage L_ξ^α remains an explicit constitutive conversion-work potential. An admissible choice of L_ξ^α is identified by the Coleman–Noll argument in Section 3.

2.12 Mobile-phase momentum equations

For the aqueous phase, the capillary contribution gives the intermediate form

$$\rho_a \mathbf{a}_a = \nabla_{\mathbf{x}} \cdot \boldsymbol{\sigma}'_a + \rho_a \mathbf{g} + \mathbf{b}_a + \phi_a \nabla_{\mathbf{x}} \lambda - \phi_a \nabla_{\mathbf{x}} \gamma_a + \sum_{\alpha} \dot{c}_a^{\alpha} (\nabla_{\mathbf{x}} \mu_a^{\alpha} - \mathbf{v}_a). \quad (66)$$

The oleic and gas equations have the same form with a replaced by o or g . The important point is that the interfacial variation contributes gradient terms $\phi_{\xi} \nabla \gamma_{\xi}$, not merely algebraic multiples of $\nabla \phi_{\xi}$. These terms must be retained until the volume-fraction pressure relations are substituted. For a mobile phase, $\lambda + \bar{p}_{\xi} - \gamma_{\xi} = 0$, so $\phi_{\xi} (\nabla \lambda - \nabla \gamma_{\xi}) = -\phi_{\xi} \nabla \bar{p}_{\xi}$. The standard phase pressure-gradient form is therefore

$$\rho_a \mathbf{a}_a = \rho_a \mathbf{g} + \mathbf{b}_a - \phi_a \nabla_{\mathbf{x}} \bar{p}_a + \sum_{\alpha} \dot{c}_a^{\alpha} (\nabla_{\mathbf{x}} \mu_a^{\alpha} - \mathbf{v}_a), \quad (67)$$

$$\rho_o \mathbf{a}_o = \rho_o \mathbf{g} + \mathbf{b}_o - \phi_o \nabla_{\mathbf{x}} \bar{p}_o + \sum_{\alpha} \dot{c}_o^{\alpha} (\nabla_{\mathbf{x}} \mu_o^{\alpha} - \mathbf{v}_o), \quad (68)$$

$$\rho_g \mathbf{a}_g = \rho_g \mathbf{g} + \mathbf{b}_g - \phi_g \nabla_{\mathbf{x}} \bar{p}_g + \sum_{\alpha} \dot{c}_g^{\alpha} (\nabla_{\mathbf{x}} \mu_g^{\alpha} - \mathbf{v}_g). \quad (69)$$

3 Constitutive Models and the Entropy Inequality

The present notation allows the solid skeleton to be treated as a single solid phase, denoted here by s , while still resolving the mineral or solid components through the mass fractions η_s^{α} . This is useful when reactions convert one solid mineral into another but the products remain part of the same load-bearing skeleton and share the same macroscopic deformation map $\mathbf{x}_s = \boldsymbol{\chi}_s(\mathbf{X}_s, t)$. In that case the solid free energy can be chosen so that the component composition only enters through composition-weighted moduli. This recovers the mixture rules used in the earlier reacting-mixture formulation, but avoids introducing a separate mechanical phase for each solid mineral.

For the present fully compressible solid model we still need the distension kinematics before it is meaningful to define an intrinsic stress. The solid phase has a macroscopic skeleton deformation $\mathbf{F}_s$, but the stress carried by a solid component is conjugate to the true, or undistended, deformation. We therefore introduce a solid distension tensor $\mathbf{A}_s$ and write

$$\mathbf{F}_s = \mathbf{A}_s \bar{\mathbf{F}}_s, \quad J_s = a_s \bar{J}_s, \quad a_s = \det \mathbf{A}_s, \quad \bar{J}_s = \det \bar{\mathbf{F}}_s. \quad (70)$$

Here $\bar{\mathbf{F}}_s$ is the true deformation seen by the load-bearing solid material, while $\mathbf{A}_s$ accounts for the change in the relation between the solid material volume and the mixture volume. In the isotropic specialization one may set $\mathbf{A}_s = a_s^{1/3} \mathbf{I}$, giving

$$\bar{\mathbf{F}}_s = a_s^{-1/3} \mathbf{F}_s. \quad (71)$$

We split the scalar distension multiplicatively into a mechanically supplied part and a reversible thermal-compositional pore-architecture factor,

$$a_s = a_s^m a_s^{\theta}(\theta, \phi_s^{\beta}), \quad a_s^{\theta} > 0. \quad (72)$$

The factor a_s^{θ} is not the intrinsic thermal expansion of the mineral grains. It is a reversible stress-free pore-architecture, or pore-volume, factor describing the temperature- and solid-volume-fraction-dependent change of pore volume, packing, and solid allocation inside the mixture volume. Thermo-poroelastic treatments of heated saturated rocks make precisely this distinction by assigning a

thermal expansion coefficient to the pore volume separately from the solid grains, so that thermal pressurization depends on the mismatch among fluid, pore-volume, and grain expansions [6, 4, 3]. Here that pore-volume part is carried by a_s^θ , while $\bar{\mathbf{F}}_s^\theta$ below carries the stress-free intrinsic thermal dilation of the true solid skeleton.

This is not the introduction of a separate distension energy. It is the kinematic device needed to separate deformation of the solid material from changes in pore-space allocation, packing, and reaction-induced filling. The evolution of a_s follows from the same mass-balance argument used by Drumheller and in the earlier reacting-mixture formulation. Summing the component material balances over solid components gives the phase mass balance

$$J_s \phi_s \bar{\rho}_s = \phi_{s0} \bar{\rho}_{s0} + \mathcal{C}_s, \quad \mathcal{C}_s = \sum_{\alpha} \mathcal{C}_s^{\alpha}. \quad (73)$$

Using $J_s = a_s \bar{J}_s$ and imposing true local conservation of mass for the solid material,

$$\bar{J}_s \bar{\rho}_s = \bar{\rho}_{s0}, \quad (74)$$

transforms (73) into

$$a_s \phi_s \bar{\rho}_{s0} = \phi_{s0} \bar{\rho}_{s0} + \mathcal{C}_s. \quad (75)$$

Thus the solid volume fraction is tied directly to the motion only when there is no mass exchange. Taking the material time derivative of $J_s = a_s \bar{J}_s$ gives

$$\frac{\dot{a}_s}{a_s} = \frac{\dot{J}_s}{J_s} - \frac{\dot{\bar{J}}_s}{\bar{J}_s}. \quad (76)$$

Since $\dot{J}_s = J_s \nabla_{\mathbf{x}} \cdot \mathbf{v}_s$ and (74) implies $\dot{\bar{J}}_s / \bar{J}_s = -\dot{\bar{\rho}}_s / \bar{\rho}_s$, the distension evolution law is

$$\frac{\dot{a}_s}{a_s} = \nabla_{\mathbf{x}} \cdot \mathbf{v}_s + \frac{\dot{\bar{\rho}}_s}{\bar{\rho}_s}. \quad (77)$$

For incompressible solid constituents, $\dot{\bar{\rho}}_s = 0$, and therefore $\dot{a}_s = a_s \nabla_{\mathbf{x}} \cdot \mathbf{v}_s$. If additionally $a_s = J_s$ in the reference state, then $a_s = J_s$ for all time; for compressible multicomponent solids, however, the intrinsic-density term makes a_s evolve independently of J_s .

For non-isothermal problems we follow the thermal split used in finite-strain thermo-hydro-mechanics, but apply it to the true deformation rather than to the total skeleton deformation. Thus

$$\bar{\mathbf{F}}_s = \bar{\mathbf{F}}_s^M \bar{\mathbf{F}}_s^\theta, \quad \bar{\mathbf{F}}_s^M = \bar{\mathbf{F}}_s (\bar{\mathbf{F}}_s^\theta)^{-1}, \quad (78)$$

where $\bar{\mathbf{F}}_s^\theta$ maps to the thermal stress-free true-solid configuration and $\bar{\mathbf{F}}_s^M$ is the mechanical part that enters the intrinsic stresses. For isotropic thermal expansion,

$$\bar{\mathbf{F}}_s^\theta = \exp \left[\int_{\theta_0}^{\theta} \alpha_{\text{sk}}(\vartheta, \phi_s^\beta) d\vartheta \right] \mathbf{I}, \quad \bar{J}_s^\theta = \exp \left[3 \int_{\theta_0}^{\theta} \alpha_{\text{sk}}(\vartheta, \phi_s^\beta) d\vartheta \right]. \quad (79)$$

Here α_{sk} is a skeleton thermal expansion coefficient that may depend on the solid composition. This is the single-skeleton analogue of assigning different thermal expansion behavior to different minerals: the composition updates α_{sk} , while all solid components remain attached to the same true deformation field.

This thermal split does not break the energy-based structure. It is a constitutive reparametrization of the free energy: instead of depending on $\bar{\mathbf{F}}_s$ directly, the elastic part of the free energy

depends on $\bar{\mathbf{F}}_s^M = \bar{\mathbf{F}}_s(\bar{\mathbf{F}}_s^\theta)^{-1}$ and on θ . The phase label remains part of the component Helmholtz energy: component α in the solid phase need not have the same free energy as component α in a liquid or gas phase. The multicomponent solid free energy is written as

$$\psi_s(\bar{\mathbf{F}}_s^M, \bar{\rho}_s, a_s, \theta, \eta_s^\beta) = \sum_\alpha \eta_s^\alpha \psi_s^\alpha(\bar{\mathbf{F}}_s^M, \bar{\rho}_s, a_s, \theta, \eta_s^\beta), \quad (80)$$

where η_s^β denotes the full list of solid component mass fractions. Temperature dependence of the free energy supplies entropy and heat capacity in the usual way, while the dependence of $\bar{\mathbf{F}}_s^\theta$ on θ and the component volume fractions ϕ_s^β encodes stress-free intrinsic thermal expansion. Any thermal or compositional dependence of scalar distension enters only through the reversible factor $a_s^\theta(\theta, \phi_s^\beta)$ in (72), not through a direct dependence on the mass fractions. For example, water in a hydrate or ice-like solid would be represented by $\psi_s^{\text{H}_2\text{O}}$, whereas liquid water would have a different function $\psi_\ell^{\text{H}_2\text{O}}$. The thermodynamic restrictions that make these constitutive choices admissible are obtained next from the entropy inequality.

3.1 Coleman–Noll procedure for the compositional thermo-reactive mixture

This subsection records the thermodynamic argument in a form that mirrors the kinematic split just introduced. Mobile phases are treated with phase-level Helmholtz potentials suitable for equation-of-state closures, while the solid phase is allowed to use component Helmholtz energies subject to density, composition, and dilation constraints.

The Coleman–Noll calculation is applied on the reduced constitutive state space. It is not a second derivation of the constrained equations of motion. Constraints that define the admissible motions and fields, such as saturation, storage, and stoichiometric source constraints, are enforced in the variational formulation and do not introduce additional Coleman–Noll coefficients unless the constrained variable is an argument of a constitutive free energy. Following the standard Liu multiplier exploitation of constrained entropy inequalities [5], we may add identically vanishing constraint-rate terms before collecting coefficients. Composition enters here because the free energies depend explicitly on η_ξ^α ; its rate is therefore restricted to the tangent space of the mass-fraction constraint,

$$\sum_\alpha \eta_\xi^\alpha = 1, \quad \sum_\alpha \dot{\eta}_\xi^\alpha = 0. \quad (81)$$

The mobile and solid phases use different Helmholtz structures. Mobile phases are modeled by phase-level equation-of-state potentials, while the solid skeleton is allowed to use a component-additive Helmholtz decomposition. For a mobile phase l , let

$$e_l = \psi_l + \theta s_l, \quad \psi_l = \psi_l(\bar{\rho}_l, \eta_l^\beta, \theta). \quad (82)$$

For each solid component let

$$e_s^\alpha = \psi_s^\alpha + \theta s_s^\alpha. \quad (83)$$

The solid phase free energy is then written as a mass-specific component mixture potential,

$$\psi_s = \sum_\alpha \eta_s^\alpha \psi_s^\alpha, \quad \rho_s \psi_s = \sum_\alpha \rho_s^\alpha \psi_s^\alpha. \quad (84)$$

The component volume fraction within the solid phase, ϕ_s^α , is a derived volume weight rather than an independent Coleman–Noll state variable. The detailed solid component mass partition is introduced below only when the stress, pressure, and distension mixture rules are read from the reversible coefficients. For the solid, the primitive component Helmholtz free energies use

the physical intrinsic density of each solid component, rather than an explicit dependence on the mixture intrinsic density $\bar{\rho}_s$. Thus the mixture density enters the Coleman–Noll calculation through the component-density parametrization $\bar{\rho}_s^\alpha = \bar{\rho}_s^\alpha(\bar{\rho}_s, \eta_s^\beta)$:

$$\mathcal{S} = \{\bar{\mathbf{F}}_s^M, a_s, \bar{\rho}_s^\beta, \theta, \eta_s^\beta\} \quad (85)$$

$$\psi_s^\alpha = \psi_s^\alpha(\bar{\mathbf{F}}_s^M, a_s, \bar{\rho}_s^\alpha, \theta, \eta_s^\beta), \quad \bar{\mathbf{F}}_s^M = \bar{\mathbf{F}}_s(\bar{\mathbf{F}}_s^\theta)^{-1}. \quad (86)$$

where $\bar{\mathbf{F}}_s^\theta = \bar{\mathbf{F}}_s^\theta(\theta, \phi_s^\beta)$ is a prescribed stress-free thermal/compositional dilation, not an independent thermodynamic state. Likewise, the isotropic distension is written as

$$a_s = a_s^m a_s^\theta(\theta, \phi_s^\beta), \quad a_s^\theta > 0, \quad (87)$$

where a_s^θ is a reversible stress-free pore-architecture, or pore-volume, thermal–compositional factor. It is distinct from the intrinsic mineral thermal dilation carried by $\bar{\mathbf{F}}_s^\theta$ and depends on composition through the solid component volume fractions ϕ_s^β , not directly through the mass fractions η_s^β . Since the balance laws still use the mixture variables $(\bar{\rho}_s, \eta_s^\beta)$, density derivatives of the solid energy are chain-ruled through $\bar{\rho}_s^\alpha(\bar{\rho}_s, \eta_s^\beta)$. In the solid specialization used below, a mixture-density change at fixed composition scales the component intrinsic densities at fixed component volume fraction:

$$\dot{\bar{\rho}}_s^\alpha = \frac{\bar{\rho}_s^\alpha}{\bar{\rho}_s} \dot{\bar{\rho}}_s + \sum_\beta \left. \frac{\partial \bar{\rho}_s^\alpha}{\partial \eta_s^\beta} \right|_{\bar{\rho}_s} \dot{\eta}_s^\beta. \quad (88)$$

Derivatives of $\bar{\mathbf{F}}_s^\theta$ with respect to composition are also total derivatives through the derived component volume fractions. Define the state sets

$$\mathcal{X} \in \{\mathcal{S}, \mathcal{F}, \mathcal{T}_s\}, \quad \mathcal{T}_s = \{\theta, \phi_s^\beta\}. \quad (89)$$

For the mobile fluid phases, we postulate the Coleman–Noll state variables

$$\mathcal{F} = \{\bar{\rho}_\ell, \eta_\ell^\beta, \theta\}. \quad (90)$$

When a derivative is written for a generic phase, let

$$\mathcal{X}_\xi = \begin{cases} \mathcal{S}, & \xi = s, \\ \mathcal{F}, & \xi = \ell, \end{cases} \quad (91)$$

We use the convention that

$$\left. \frac{\partial \Gamma_\xi}{\partial z} \right|_{\mathcal{X}} \quad (92)$$

denotes differentiation with respect to z while holding fixed all other members of the state set $\mathcal{X}$. We write

$$\left. \frac{\partial \bar{\mathbf{F}}_s^\theta}{\partial \eta_s^\beta} \right|_{\mathcal{S}} = \sum_\gamma \left. \frac{\partial \bar{\mathbf{F}}_s^\theta}{\partial \phi_s^\gamma} \right|_{\mathcal{T}_s} \left. \frac{\partial \phi_s^\gamma}{\partial \eta_s^\beta} \right|_{\mathcal{S}}. \quad (93)$$

The corresponding scalar distension rate is most cleanly written in logarithmic form:

$$\begin{aligned} \frac{\dot{a}_s}{a_s} &= \frac{\dot{a}_s^m}{a_s^m} + \frac{\dot{a}_s^\theta}{a_s^\theta} \\ &= \frac{\dot{a}_s^m}{a_s^m} + \left. \frac{\partial \ln a_s^\theta}{\partial \theta} \right|_{\phi_s^\gamma} \dot{\theta} + \sum_\beta \left[\sum_\gamma \left. \frac{\partial \ln a_s^\theta}{\partial \phi_s^\gamma} \right|_{\mathcal{T}_s} \left. \frac{\partial \phi_s^\gamma}{\partial \eta_s^\beta} \right|_{\mathcal{S}} \right] \dot{\eta}_s^\beta. \end{aligned} \quad (94)$$

Here $\dot{a}_s^m/a_s^m$ denotes the logarithmic distension rate supplied by the mechanical, storage, or reaction kinematics at fixed temperature and composition. If a model chooses no explicit pore-architecture thermal or compositional distension, the last two terms vanish and the earlier rate collection is recovered. Thus the fluid-phase free-energy rates entering the entropy inequality are

$$\sum_{\ell} \rho_{\ell} \dot{\psi}_{\ell} = \sum_{\ell} \rho_{\ell} \left(\left. \frac{\partial \psi_{\ell}}{\partial \bar{\rho}_{\ell}} \right|_{\mathcal{F}} \dot{\bar{\rho}}_{\ell} + \sum_{\beta} \left. \frac{\partial \psi_{\ell}}{\partial \eta_{\ell}^{\beta}} \right|_{\mathcal{F}} \dot{\eta}_{\ell}^{\beta} + \left. \frac{\partial \psi_{\ell}}{\partial \theta} \right|_{\mathcal{F}} \dot{\theta} \right). \quad (95)$$

The dependent quantities

$$\begin{aligned} \rho_{\xi}^{\alpha} &= \phi_{\xi} \bar{\rho}_{\xi} \eta_{\xi}^{\alpha}, \\ \sum_{\alpha} \eta_{\xi}^{\alpha} &= 1, \\ \bar{J}_s &= \det \bar{\mathbf{F}}_s \\ &= \frac{J_s}{a_s} \end{aligned} \quad (96)$$

For the solid, component additivity gives

$$\begin{aligned} \sum_{\alpha} \rho_s^{\alpha} \dot{\psi}_s^{\alpha} &= \rho_s \sum_{\alpha} \eta_s^{\alpha} \dot{\psi}_s^{\alpha} \\ &= \rho_s \sum_{\alpha} \eta_s^{\alpha} \left(\left. \frac{\partial \psi_s^{\alpha}}{\partial \bar{\mathbf{F}}_s^M} \right|_S : \dot{\bar{\mathbf{F}}}_s^M + a_s \left. \frac{\partial \psi_s^{\alpha}}{\partial a_s} \right|_S \frac{\dot{a}_s^m}{a_s^m} + a_s \left. \frac{\partial \psi_s^{\alpha}}{\partial a_s} \right|_S \left. \frac{\partial \ln a_s^{\theta}}{\partial \theta} \right|_{\phi_s^{\gamma}} \dot{\theta} \\ &\quad + \sum_{\beta} a_s \left. \frac{\partial \psi_s^{\alpha}}{\partial a_s} \right|_S \left[\sum_{\gamma} \left. \frac{\partial \ln a_s^{\theta}}{\partial \phi_s^{\gamma}} \right|_{\mathcal{T}_s} \frac{\partial \phi_s^{\gamma}}{\partial \eta_s^{\beta}} \right] \dot{\eta}_s^{\beta} \\ &\quad + \left. \frac{\partial \psi_s^{\alpha}}{\partial \bar{\rho}_s^{\alpha}} \right|_S \left[\frac{\bar{\rho}_s^{\alpha}}{\bar{\rho}_s} \dot{\bar{\rho}}_s + \sum_{\beta} \left. \frac{\partial \bar{\rho}_s^{\alpha}}{\partial \eta_s^{\beta}} \right|_{\bar{\rho}_s} \dot{\eta}_s^{\beta} \right] + \sum_{\beta} \left. \frac{\partial \psi_s^{\alpha}}{\partial \eta_s^{\beta}} \right|_S \dot{\eta}_s^{\beta} \\ &\quad + \left. \frac{\partial \psi_s^{\alpha}}{\partial \theta} \right|_S \dot{\theta} \right). \end{aligned} \quad (97)$$

The Coleman–Noll argument treats $\dot{\bar{\mathbf{F}}}_s$ as the independent solid deformation rate. The mechanical deformation gradient is a dependent constitutive argument, so substituting the prescribed thermal/compositional stress-free dilation rate gives

$$\begin{aligned} \dot{\bar{\mathbf{F}}}_s^M &= \dot{\bar{\mathbf{F}}}_s (\bar{\mathbf{F}}_s^{\theta})^{-1} - \bar{\mathbf{F}}_s^M \left. \frac{\partial \bar{\mathbf{F}}_s^{\theta}}{\partial \theta} \right|_{\mathcal{T}_s} (\bar{\mathbf{F}}_s^{\theta})^{-1} \dot{\theta} \\ &\quad - \sum_{\beta} \bar{\mathbf{F}}_s^M \left. \frac{\partial \bar{\mathbf{F}}_s^{\theta}}{\partial \eta_s^{\beta}} \right|_S (\bar{\mathbf{F}}_s^{\theta})^{-1} \dot{\eta}_s^{\beta}. \end{aligned} \quad (98)$$

It is useful to define the tensor conjugate to the stress-free dilation,

$$\mathbf{H}_s^{\alpha} = (\bar{\mathbf{F}}_s^M)^T \left. \frac{\partial \psi_s^{\alpha}}{\partial \bar{\mathbf{F}}_s^M} \right|_S (\bar{\mathbf{F}}_s^{\theta})^{-T}. \quad (99)$$

It is also useful to name the reversible distension-force residual

$$\mathcal{D}_s = \frac{1}{3} \text{tr}(\boldsymbol{\sigma}'_s) - \rho_s a_s \sum_{\alpha} \eta_s^{\alpha} \left. \frac{\partial \psi_s^{\alpha}}{\partial a_s} \right|_S. \quad (100)$$

The mobile-phase Helmholtz energies are phase EOS potentials using the fluid-state variables in (90). They are not assumed to decompose into mass-fraction averages of component Helmholtz energies. We assign no elastic stress power to the mobile phases, so

$$\boldsymbol{\sigma}'_\ell = \mathbf{0}. \quad (101)$$

The reversible interfacial energy used in the variational formulation must also be included in the entropy bookkeeping. For the mobile phases,

$$\psi_\gamma^{\text{vol}} = \phi\gamma(S_a, S_g), \quad \dot{\psi}_\gamma^{\text{vol}} = \sum_l \gamma_l \dot{\phi}_l, \quad (102)$$

where $\gamma_l = \partial(\phi\gamma)/\partial\phi_l$ are the primitive interfacial potentials introduced in (35). The volume fractions are constrained by $\sum_\xi \phi_\xi = 1$, so their rates are not independent. When the capillary Helmholtz rate is retained explicitly, we add the identically vanishing saturation-constraint rate $\lambda \sum_\xi \dot{\phi}_\xi$ and retain the corresponding $\dot{\phi}_\xi/\phi_\xi$ contribution from the component-storage multiplier. This capillary Helmholtz energy is taken to be reversible and independent of temperature in the present model. If γ were temperature dependent, the corresponding interfacial entropy term would have to be retained as well. Define the source-carried specific potentials and entropies by

$$\widehat{\psi}_\xi^\alpha = \begin{cases} \psi_l, & \xi = l \text{ mobile}, \\ \psi_s^\alpha, & \xi = s, \end{cases} \quad \widehat{s}_\xi^\alpha = \begin{cases} s_l, & \xi = l \text{ mobile}, \\ s_s^\alpha, & \xi = s. \end{cases} \quad (103)$$

Equivalently, the source-carried internal energy is $\widehat{e}_\xi^\alpha = \widehat{\psi}_\xi^\alpha + \theta \widehat{s}_\xi^\alpha$. This notation keeps the open-system source terms phase/component-unified: mobile source mass carries the mobile phase thermodynamic state, while solid source mass carries the corresponding solid component state. The energy balance is written in phase form. The phase internal-energy rate is left as $\rho_\xi \dot{e}_\xi$ here; the mobile and solid Helmholtz substitutions are made below when collecting the entropy inequality:

$$\rho_\xi \dot{e}_\xi + \sum_\alpha \dot{c}_\xi^\alpha \widehat{e}_\xi^\alpha = \boldsymbol{\sigma}'_\xi : \mathbf{L}_\xi + \phi_\xi \bar{p}_\xi \frac{\dot{\rho}_\xi}{\rho_\xi} + \sum_\alpha \frac{\tau_\xi^\alpha}{\phi_\xi} \dot{\phi}_\xi + \sum_\alpha \frac{\tau_\xi^\alpha}{\eta_\xi^\alpha} \dot{\eta}_\xi^\alpha - \sum_\alpha L_\xi^\alpha \dot{c}_\xi^\alpha - \nabla_{\mathbf{x}} \cdot \mathbf{q}_\xi + \rho_\xi r_\xi + \rho_\xi \dot{e}_\xi^{\text{int}}. \quad (104)$$

The storage-multiplier terms are the volume-fraction and compositional analogs of the pressure power. The multipliers τ_ξ^α enforce component material storage in (19); when the admissible rates $\dot{\phi}_\xi$ and $\dot{\eta}_\xi^\alpha$ are retained, they do work against the normalized storage multipliers τ_ξ^α/ϕ_ξ and $\tau_\xi^\alpha/\eta_\xi^\alpha$. The coefficient L_ξ^α is the same component conversion-work potential introduced in (17); the sign convention here is chosen so that a source $\dot{c}_\xi^\alpha$ carries the free-energy cost $\widehat{\psi}_\xi^\alpha + L_\xi^\alpha$. For the solid phase, the mechanical power in (104) must be decomposed with the same kinematic split used in the constitutive law. In the isotropic dilation specialization,

$$\mathbf{F}_s = a_s^{1/3} \bar{\mathbf{F}}_s, \quad (105)$$

so

$$\mathbf{L}_s = \dot{\mathbf{F}}_s \mathbf{F}_s^{-1} = \frac{1}{3} \frac{\dot{a}_s}{a_s} \mathbf{I} + \dot{\bar{\mathbf{F}}}_s \bar{\mathbf{F}}_s^{-1}. \quad (106)$$

Therefore

$$\boldsymbol{\sigma}'_s : \mathbf{L}_s = \left[\boldsymbol{\sigma}'_s \bar{\mathbf{F}}_s^{-T} \right] : \dot{\bar{\mathbf{F}}}_s + \frac{1}{3a_s} \text{tr}(\boldsymbol{\sigma}'_s) \dot{a}_s. \quad (107)$$

The entropy inequality is

$$\begin{aligned}
0 \leq & \sum_l \left[\rho_l \dot{s}_l + \sum_\alpha \dot{c}_l^\alpha s_l + \nabla_{\mathbf{x}} \cdot \left(\frac{\mathbf{q}_l}{\theta} \right) - \frac{\rho_l r_l}{\theta} \right] \\
& + \sum_\alpha (\rho_s^\alpha \dot{s}_s^\alpha + \dot{c}_s^\alpha s_s^\alpha) + \nabla_{\mathbf{x}} \cdot \left(\frac{\mathbf{q}_s}{\theta} \right) - \frac{\rho_s r_s}{\theta}.
\end{aligned} \tag{108}$$

Using (104) to eliminate $\rho_\xi r_\xi$ from (108), and then using the Legendre relations (82) and (83), gives the reduced entropy inequality. At this stage we also add the identically vanishing rate of the composition-normalization constraint, $\pi_\xi \sum_\alpha \dot{\eta}_\xi^\alpha$, and the identically vanishing saturation-constraint rate $\lambda \sum_\xi \dot{\phi}_\xi$, so that the Coleman–Noll coefficient collection is performed on the constrained tangent spaces.

$$\begin{aligned}
0 \leq & \sum_l \left[\frac{1}{\theta} \left(\boldsymbol{\sigma}'_l : \mathbf{L}_l + \phi_l \bar{p}_l \frac{\dot{\bar{\rho}}_l}{\bar{\rho}_l} + \sum_\alpha \frac{\tau_l^\alpha}{\phi_l} \dot{\phi}_l \right. \right. \\
& + \sum_\alpha \frac{\tau_l^\alpha}{\eta_l^\alpha} \dot{\eta}_l^\alpha + \pi_l \sum_\alpha \dot{\eta}_l^\alpha - \rho_l (\dot{\psi}_l + s_l \dot{\theta}) \\
& \left. \left. - \sum_\alpha \dot{c}_l^\alpha (\hat{\psi}_l^\alpha + L_l^\alpha) + \rho_l e_l^{\text{int}} \right) - \frac{\mathbf{q}_l \cdot \nabla_{\mathbf{x}} \theta}{\theta^2} \right] \\
& + \frac{1}{\theta} \left(\boldsymbol{\sigma}'_s : \mathbf{L}_s + \phi_s \bar{p}_s \frac{\dot{\bar{\rho}}_s}{\bar{\rho}_s} + \sum_\alpha \frac{\tau_s^\alpha}{\phi_s} \dot{\phi}_s \right. \\
& + \sum_\alpha \frac{\tau_s^\alpha}{\eta_s^\alpha} \dot{\eta}_s^\alpha + \pi_s \sum_\alpha \dot{\eta}_s^\alpha - \sum_\alpha \rho_s^\alpha (\dot{\psi}_s^\alpha + s_s^\alpha \dot{\theta}) \\
& \left. - \sum_\alpha \dot{c}_s^\alpha (\hat{\psi}_s^\alpha + L_s^\alpha) + \rho_s e_s^{\text{int}} \right) - \frac{\mathbf{q}_s \cdot \nabla_{\mathbf{x}} \theta}{\theta^2} \\
& + \frac{1}{\theta} \left(\lambda \sum_\xi \dot{\phi}_\xi - \sum_l \gamma_l \dot{\phi}_l \right).
\end{aligned} \tag{109}$$

Substituting (97), (94), (98), (107), and (95) gives the collected solid–fluid contribution.

$$\begin{aligned}
0 \leq & \frac{1}{\theta} \left\{ \left[\boldsymbol{\sigma}'_s \bar{\mathbf{F}}_s^{-T} - \rho_s \sum_{\alpha} \eta_s^{\alpha} \frac{\partial \psi_s^{\alpha}}{\partial \bar{\mathbf{F}}_s^M} \Big|_{\mathcal{S}} (\bar{\mathbf{F}}_s^{\theta})^{-T} \right] : \dot{\bar{\mathbf{F}}}_s \right. \\
& + \mathcal{D}_s \frac{\dot{a}_s^m}{a_s^m} \\
& + \left[\frac{\phi_s \bar{p}_s}{\bar{\rho}_s} - \rho_s \sum_{\alpha} \eta_s^{\alpha} \frac{\partial \psi_s^{\alpha}}{\partial \bar{\rho}_s^{\alpha}} \Big|_{\mathcal{S}} \frac{\bar{\rho}_s^{\alpha}}{\bar{\rho}_s} \right] \dot{\bar{\rho}}_s + \sum_l \left[\frac{\phi_l \bar{p}_l}{\bar{\rho}_l} - \rho_l \frac{\partial \psi_l}{\partial \bar{\rho}_l} \Big|_{\mathcal{F}} \right] \dot{\bar{\rho}}_l \\
& + \left[- \sum_l \rho_l \left(s_l + \frac{\partial \psi_l}{\partial \theta} \Big|_{\mathcal{F}} \right) - \rho_s \sum_{\alpha} \eta_s^{\alpha} \left(s_s^{\alpha} + \frac{\partial \psi_s^{\alpha}}{\partial \theta} \Big|_{\mathcal{S}} \right) + \rho_s \sum_{\alpha} \eta_s^{\alpha} \mathbf{H}_s^{\alpha} : \frac{\partial \bar{\mathbf{F}}_s^{\theta}}{\partial \theta} \Big|_{\mathcal{T}_s} + \mathcal{D}_s \frac{\partial \ln a_s^{\theta}}{\partial \theta} \Big|_{\phi_s^{\gamma}} \right] \dot{\theta} \\
& - \sum_l \rho_l \sum_{\beta} \frac{\partial \psi_l}{\partial \eta_l^{\beta}} \Big|_{\mathcal{F}} \dot{\eta}_l^{\beta} \\
& + \rho_s \sum_{\beta} \left[- \frac{\partial \psi_s}{\partial \eta_s^{\beta}} \Big|_{\mathcal{S}} - \sum_{\alpha} \eta_s^{\alpha} \frac{\partial \psi_s^{\alpha}}{\partial \bar{\rho}_s^{\alpha}} \Big|_{\mathcal{S}} \frac{\partial \bar{\rho}_s^{\alpha}}{\partial \eta_s^{\beta}} \Big|_{\bar{\rho}_s} \right. \\
& \quad \left. + \sum_{\alpha} \eta_s^{\alpha} \mathbf{H}_s^{\alpha} : \frac{\partial \bar{\mathbf{F}}_s^{\theta}}{\partial \eta_s^{\beta}} \Big|_{\mathcal{S}} + \frac{\mathcal{D}_s}{\rho_s} \left[\sum_{\gamma} \frac{\partial \ln a_s^{\theta}}{\partial \phi_s^{\gamma}} \Big|_{\mathcal{T}_s} \frac{\partial \phi_s^{\gamma}}{\partial \eta_s^{\beta}} \Big|_{\mathcal{S}} \right] \right] \dot{\eta}_s^{\beta} \\
& + \sum_{\xi} \sum_{\beta} \left(\frac{\tau_{\xi}^{\beta}}{\eta_{\xi}^{\beta}} + \pi_{\xi} \right) \dot{\eta}_{\xi}^{\beta} \\
& - \sum_m \left(\sum_{\xi} \sum_{\alpha} \nu_{\xi m}^{\alpha} (\hat{\psi}_{\xi}^{\alpha} + L_{\xi}^{\alpha}) \right) \dot{r}_m \\
& + \sum_{\xi} \left(\lambda + \sum_{\alpha} \frac{\tau_{\xi}^{\alpha}}{\phi_{\xi}} \right) \dot{\phi}_{\xi} - \sum_l \gamma_l \dot{\phi}_l \\
& \left. + \sum_{\xi} \rho_{\xi} e_{\xi}^{\text{int}} - \sum_{\xi} \frac{\mathbf{q}_{\xi} \cdot \nabla_{\mathbf{x}} \theta}{\theta} \right\}. \tag{110}
\end{aligned}$$

The capillary term is not a new dissipative mechanism. Since the volume-fraction rates are constrained, the mobile-phase coefficient is the reversible condition

$$\begin{aligned}
0 &= \lambda + \sum_{\alpha} \frac{\tau_l^{\alpha}}{\phi_l} - \gamma_l \\
&= \lambda + \bar{p}_l - \gamma_l. \tag{111}
\end{aligned}$$

Thus the interfacial energy supplies the capillary pressure construction and equivalent pressure relation, while leaving no residual entropy production for equilibrium capillarity. A dynamic or hysteretic capillary law would instead leave the uncanceled part of the interfacial Helmholtz rate as a capillary dissipation term. The phase derivative with respect to composition is the derivative of the corresponding phase Helmholtz potential. For mobile phases no elastic stress-free dilation term appears, so the composition coefficient gives the same stationarity condition obtained from Hamilton's principle in (59):

$$0 = \frac{\tau_{\ell}^{\beta}}{\eta_{\ell}^{\beta}} + \pi_{\ell} - \rho_{\ell} \frac{\partial \psi_{\ell}}{\partial \eta_{\ell}^{\beta}} \Big|_{\mathcal{F}}, \quad \beta = 1, \dots, N. \tag{112}$$

Equivalently,

$$\rho_\ell \left. \frac{\partial \psi_\ell}{\partial \eta_\ell^\beta} \right|_{\mathcal{F}} - \frac{\tau_\ell^\beta}{\eta_\ell^\beta} = \pi_\ell. \quad (113)$$

For the solid phase, the same coefficient collection contains both the component-density chain rule and the explicit stress-free compositional dilation correction from (98). If the reversible pore architecture factor $a_s^\theta(\theta, \phi_s^\gamma)$ depends on composition, the analogous distension chain-rule term passes through the component volume fractions:

$$0 = \frac{\tau_s^\beta}{\eta_s^\beta} + \pi_s - \rho_s \left[\left. \frac{\partial \psi_s}{\partial \eta_s^\beta} \right|_{\mathcal{S}} + \sum_\alpha \eta_s^\alpha \left. \frac{\partial \psi_s^\alpha}{\partial \bar{\rho}_s^\alpha} \right|_{\mathcal{S}} \left. \frac{\partial \bar{\rho}_s^\alpha}{\partial \eta_s^\beta} \right|_{\bar{\rho}_s} \right] + \rho_s \sum_\alpha \eta_s^\alpha \mathbf{H}_s^\alpha : \left. \frac{\partial \bar{\mathbf{F}}_s^\theta}{\partial \eta_s^\beta} \right|_{\mathcal{S}} + \mathcal{D}_s \left[\sum_\gamma \left. \frac{\partial \ln a_s^\theta}{\partial \phi_s^\gamma} \right|_{\mathcal{T}_s} \left. \frac{\partial \phi_s^\gamma}{\partial \eta_s^\beta} \right|_{\mathcal{S}} \right], \quad \beta = 1, \dots, N. \quad (114)$$

Eliminating π_ℓ gives the mobile-phase projected restriction on the tangent space,

$$\rho_\ell \left(\left. \frac{\partial \psi_\ell}{\partial \eta_\ell^\beta} \right|_{\mathcal{F}} - \left. \frac{\partial \psi_\ell}{\partial \eta_\ell^N} \right|_{\mathcal{F}} \right) = \frac{\tau_\ell^\beta}{\eta_\ell^\beta} - \frac{\tau_\ell^N}{\eta_\ell^N}, \quad \beta = 1, \dots, N-1. \quad (115)$$

For the solid, eliminating π_s gives the corresponding projected restriction with the differences of the density-chain and dilation corrections retained:

$$\begin{aligned} & \rho_s \left(\left. \frac{\partial \psi_s}{\partial \eta_s^\beta} \right|_{\mathcal{S}} - \left. \frac{\partial \psi_s}{\partial \eta_s^N} \right|_{\mathcal{S}} \right) + \rho_s \sum_\alpha \eta_s^\alpha \left. \frac{\partial \psi_s^\alpha}{\partial \bar{\rho}_s^\alpha} \right|_{\mathcal{S}} \left(\left. \frac{\partial \bar{\rho}_s^\alpha}{\partial \eta_s^\beta} \right|_{\bar{\rho}_s} - \left. \frac{\partial \bar{\rho}_s^\alpha}{\partial \eta_s^N} \right|_{\bar{\rho}_s} \right) \\ & - \rho_s \sum_\alpha \eta_s^\alpha \mathbf{H}_s^\alpha : \left(\left. \frac{\partial \bar{\mathbf{F}}_s^\theta}{\partial \eta_s^\beta} \right|_{\mathcal{S}} - \left. \frac{\partial \bar{\mathbf{F}}_s^\theta}{\partial \eta_s^N} \right|_{\mathcal{S}} \right) \\ & - \mathcal{D}_s \left(\sum_\gamma \left. \frac{\partial \ln a_s^\theta}{\partial \phi_s^\gamma} \right|_{\mathcal{T}_s} \left[\left. \frac{\partial \phi_s^\gamma}{\partial \eta_s^\beta} \right|_{\mathcal{S}} - \left. \frac{\partial \phi_s^\gamma}{\partial \eta_s^N} \right|_{\mathcal{S}} \right] \right) \\ & = \frac{\tau_s^\beta}{\eta_s^\beta} - \frac{\tau_s^N}{\eta_s^N}, \quad \beta = 1, \dots, N-1. \end{aligned} \quad (116)$$

The deformation coefficient gives

$$\boldsymbol{\sigma}'_s = \rho_s \sum_\alpha \eta_s^\alpha \left. \frac{\partial \psi_s^\alpha}{\partial \bar{\mathbf{F}}_s^M} \right|_{\mathcal{S}} (\bar{\mathbf{F}}_s^M)^T = \phi_s \sum_\alpha \phi_s^\alpha \bar{\rho}_s^\alpha \left. \frac{\partial \psi_s^\alpha}{\partial \bar{\mathbf{F}}_s^M} \right|_{\mathcal{S}} (\bar{\mathbf{F}}_s^M)^T. \quad (117)$$

The component partial Cauchy stress uses the current component partial density. The intrinsic component stress is the corresponding normalized stress within the solid phase:

$$\boldsymbol{\sigma}'_s{}^\alpha = \rho_s^\alpha \left. \frac{\partial \psi_s^\alpha}{\partial \bar{\mathbf{F}}_s^M} \right|_{\mathcal{S}} (\bar{\mathbf{F}}_s^M)^T = \phi_s \phi_s^\alpha \bar{\sigma}'_s{}^\alpha, \quad \bar{\sigma}'_s{}^\alpha = \bar{\rho}_s^\alpha \left. \frac{\partial \psi_s^\alpha}{\partial \bar{\mathbf{F}}_s^M} \right|_{\mathcal{S}} (\bar{\mathbf{F}}_s^M)^T. \quad (118)$$

Thus

$$\boldsymbol{\sigma}'_s = \sum_\alpha \boldsymbol{\sigma}'_s{}^\alpha = \phi_s \sum_\alpha \phi_s^\alpha \bar{\sigma}'_s{}^\alpha. \quad (119)$$

Equivalently, if $\rho_{s0}^\alpha \psi_s^\alpha$ is the energy per component reference mixture volume, then

$$\mathbf{P}'_s{}^\alpha = \rho_{s0}^\alpha \left. \frac{\partial \psi_s^\alpha}{\partial \bar{\mathbf{F}}_s^M} \right|_{\mathcal{S}} = J_s \boldsymbol{\sigma}'_s{}^\alpha (\bar{\mathbf{F}}_s^M)^{-T} \quad (120)$$

is the component first Piola–Kirchhoff stress. The distension coefficient gives

$$\frac{1}{3} \text{tr}(\boldsymbol{\sigma}'_s) = \rho_s a_s \sum_{\alpha} \eta_s^{\alpha} \left. \frac{\partial \psi_s^{\alpha}}{\partial a_s} \right|_{\mathcal{S}}, \quad (121)$$

The solid pressure relation follows from the same coefficient collection. The component mass identity gives

$$\rho_s^{\alpha} = \phi_s \bar{\rho}_s \eta_s^{\alpha} = \phi_s \phi_s^{\alpha} \bar{\rho}_s^{\alpha}, \quad \eta_s^{\alpha} \bar{\rho}_s = \phi_s^{\alpha} \bar{\rho}_s^{\alpha}. \quad (122)$$

Since ψ_s^{α} depends on the component intrinsic density rather than directly on $\bar{\rho}_s$, the part of the solid Helmholtz rate proportional to $\dot{\bar{\rho}}_s$ is

$$\rho_s \sum_{\alpha} \eta_s^{\alpha} \left. \frac{\partial \psi_s^{\alpha}}{\partial \bar{\rho}_s^{\alpha}} \right|_{\mathcal{S}} \frac{\bar{\rho}_s^{\alpha}}{\bar{\rho}_s} \dot{\bar{\rho}}_s. \quad (123)$$

The pressure-power term contributes $(\phi_s \bar{p}_s / \bar{\rho}_s) \dot{\bar{\rho}}_s$. Therefore the coefficient of the independent rate $\dot{\bar{\rho}}_s$ is

$$0 = \frac{\phi_s \bar{p}_s}{\bar{\rho}_s} - \rho_s \sum_{\alpha} \eta_s^{\alpha} \left. \frac{\partial \psi_s^{\alpha}}{\partial \bar{\rho}_s^{\alpha}} \right|_{\mathcal{S}} \frac{\bar{\rho}_s^{\alpha}}{\bar{\rho}_s}. \quad (124)$$

Using $\rho_s = \phi_s \bar{\rho}_s$ and $\eta_s^{\alpha} \bar{\rho}_s = \phi_s^{\alpha} \bar{\rho}_s^{\alpha}$ gives

$$\bar{p}_s = \sum_{\alpha} \phi_s^{\alpha} (\bar{\rho}_s^{\alpha})^2 \left. \frac{\partial \psi_s^{\alpha}}{\partial \bar{\rho}_s^{\alpha}} \right|_{\mathcal{S}}. \quad (125)$$

Thus, defining each component intrinsic pressure by

$$\bar{p}_s^{\alpha} = (\bar{\rho}_s^{\alpha})^2 \left. \frac{\partial \psi_s^{\alpha}}{\partial \bar{\rho}_s^{\alpha}} \right|_{\mathcal{S}}, \quad (126)$$

the pressure coefficients give

$$\begin{aligned} \bar{p}_s &= \sum_{\alpha} \phi_s^{\alpha} \bar{p}_s^{\alpha}, \quad \bar{p}_s^{\alpha} = (\bar{\rho}_s^{\alpha})^2 \left. \frac{\partial \psi_s^{\alpha}}{\partial \bar{\rho}_s^{\alpha}} \right|_{\mathcal{S}}, \\ \bar{p}_l &= (\bar{\rho}_l)^2 \left. \frac{\partial \psi_l}{\partial \bar{\rho}_l} \right|_{\mathcal{F}}. \end{aligned} \quad (127)$$

The entropy coefficient gives

$$\begin{aligned} s_l &= - \left. \frac{\partial \psi_l}{\partial \theta} \right|_{\mathcal{F}}, \quad l \text{ mobile}, \\ \sum_{\alpha} \eta_s^{\alpha} s_s^{\alpha} &= - \sum_{\alpha} \eta_s^{\alpha} \left. \frac{\partial \psi_s^{\alpha}}{\partial \theta} \right|_{\mathcal{S}} + \sum_{\alpha} \eta_s^{\alpha} \mathbf{H}_s^{\alpha} : \left. \frac{\partial \bar{\mathbf{F}}_s^{\theta}}{\partial \theta} \right|_{\mathcal{T}_s} + \frac{\mathcal{D}_s}{\rho_s} \left. \frac{\partial \ln a_s^{\theta}}{\partial \theta} \right|_{\phi_s^{\gamma}}. \end{aligned} \quad (128)$$

The solid composition restrictions above are the place where component-density parametrization, stress-free compositional dilation, and pore-architecture distension modify the usual composition coefficient. When $\dot{a}_s^m / a_s^m$ is an admissible independent logarithmic distension rate, the distension coefficient (121) makes the explicit a_s^{θ} chain-rule terms vanish from the reduced entropy and composition restrictions; if a model prescribes the reversible pore-architecture factor, those displayed chain-rule terms are the corresponding reversible generalized forces.

Finally, using $\dot{c}_\xi^\alpha = \sum_m \nu_{\xi m}^\alpha \dot{r}_m$, the source term in (110) becomes

$$\begin{aligned} & - \sum_\xi \sum_\alpha \dot{c}_\xi^\alpha (\hat{\psi}_\xi^\alpha + L_\xi^\alpha) \\ & = - \sum_m \left[\sum_\xi \sum_\alpha \nu_{\xi m}^\alpha (\hat{\psi}_\xi^\alpha + L_\xi^\alpha) \right] \dot{r}_m. \end{aligned} \quad (129)$$

This term is the open-system source contribution to the entropy inequality. To make the entropy production use the same component conversion potential that appears in the variational stoichiometric constraint, we choose

$$L_\xi^\alpha = \mu_\xi^\alpha - \hat{\psi}_\xi^\alpha. \quad (130)$$

Then $\hat{\psi}_\xi^\alpha + L_\xi^\alpha = \mu_\xi^\alpha$ for every phase/component source. The source projection is

$$- \sum_\xi \sum_\alpha \dot{c}_\xi^\alpha \mu_\xi^\alpha = - \sum_m \left(\sum_\xi \sum_\alpha \nu_{\xi m}^\alpha \mu_\xi^\alpha \right) \dot{r}_m. \quad (131)$$

The mechanism affinity identified by the variational conversion constraint is

$$\mathcal{A}_m = - \sum_\xi \sum_\alpha \mu_\xi^\alpha \nu_{\xi m}^\alpha. \quad (132)$$

Thus the source projection (131) is exactly the mechanism force power. After the reversible coefficients have been set to zero, the remaining inequality can therefore be written as

$$0 \leq \sum_m \mathcal{A}_m \dot{r}_m + \sum_\xi \rho_\xi e_\xi^{\text{int}} - \sum_\xi \frac{\mathbf{q}_\xi \cdot \nabla_{\mathbf{x}} \theta}{\theta}. \quad (133)$$

Admissible reaction kinetics must make $\sum_m \mathcal{A}_m \dot{r}_m \geq 0$, for example through a closure $\dot{r}_m = \mathcal{K}_m(\mathcal{A}_m, \text{state})$ with $\mathcal{A}_m \dot{r}_m \geq 0$ for each independent mechanism.

The interphase exchange term can be closed by the same quadratic argument used for the three-phase reacting mixture. Following Bedford and Drumheller [2], conservation of energy for the entire mixture requires the internal interaction-energy supplies and the interphase momentum supplies to satisfy

$$\sum_\xi \left(\rho_\xi e_\xi^{\text{int}} + \mathbf{b}_\xi \cdot \mathbf{v}_\xi \right) = 0. \quad (134)$$

Here $\mathbf{b}_\xi$ is the net interphase momentum supply appearing in the phase momentum balance. Decompose it into pairwise interaction forces,

$$\mathbf{b}_\xi = \sum_{\zeta \neq \xi} \mathbf{b}_{\zeta \xi}, \quad \mathbf{b}_{\zeta \xi} = -\mathbf{b}_{\xi \zeta}, \quad (135)$$

where $\mathbf{b}_{\zeta \xi}$ is the force exerted on phase ξ by phase ζ . For each unordered pair, let $\mathbf{u}_{\xi \zeta}$ denote the pairwise relative velocity,

$$\mathbf{u}_{\xi \zeta} = \mathbf{v}_\xi - \mathbf{v}_\zeta. \quad (136)$$

This velocity difference is distinct from mass-weighted Darcy or diffusion fluxes, which can be introduced later by multiplying a relative velocity by the appropriate partial density or mobility. The energy pairing remains simplest in terms of $\mathbf{u}_{\xi \zeta}$. Then (134) gives

$$\sum_\xi \rho_\xi e_\xi^{\text{int}} = - \sum_{\xi < \zeta} \mathbf{b}_{\zeta \xi} \cdot \mathbf{u}_{\xi \zeta}. \quad (137)$$

A linear dissipative interaction law may be written as

$$\mathbf{b}_{\zeta\xi} = -\mathbf{k}_{\xi\zeta}^{-1}\mathbf{u}_{\xi\zeta}, \quad \mathbf{k}_{\xi\zeta} = \mathbf{k}_{\xi\zeta}^T, \quad \mathbf{y} \cdot \mathbf{k}_{\xi\zeta}\mathbf{y} > 0 \quad \forall \mathbf{y} \neq \mathbf{0}. \quad (138)$$

Substitution into (137) gives

$$\sum_{\xi} \rho_{\xi} e_{\xi}^{\text{int}} = \sum_{\xi < \zeta} \mathbf{u}_{\xi\zeta} \cdot \mathbf{k}_{\xi\zeta}^{-1} \mathbf{u}_{\xi\zeta} \geq 0. \quad (139)$$

Thus interphase momentum exchange contributes nonnegative entropy production whenever the mobility tensors are positive definite.

The heat-conduction term is made admissible by a Fourier-type closure

$$\mathbf{q}_{\xi} = -\mathbf{K}_{\xi} \nabla_{\mathbf{x}} \theta, \quad \mathbf{K}_{\xi} = \mathbf{K}_{\xi}^T, \quad \mathbf{y} \cdot \mathbf{K}_{\xi} \mathbf{y} \geq 0, \quad (140)$$

with $\theta > 0$. Then

$$-\sum_{\xi} \frac{\mathbf{q}_{\xi} \cdot \nabla_{\mathbf{x}} \theta}{\theta} = \sum_{\xi} \frac{\nabla_{\mathbf{x}} \theta \cdot \mathbf{K}_{\xi} \nabla_{\mathbf{x}} \theta}{\theta} \geq 0. \quad (141)$$

Equations (139) and (141) are sufficient to make (133) thermodynamically admissible.

3.2 Solid mixture rules and nonlinear Biot coefficient

The Coleman–Noll restrictions above close the solid constitutive structure in terms of the component Helmholtz energies. For the single load-bearing skeleton, the component volume fractions, mass fractions, and intrinsic densities satisfy

$$\bar{\rho}_s = \sum_{\alpha} \phi_s^{\alpha} \bar{\rho}_s^{\alpha}, \quad (142a)$$

$$\eta_s^{\alpha} = \frac{\phi_s^{\alpha} \bar{\rho}_s^{\alpha}}{\bar{\rho}_s}, \quad (142b)$$

$$\phi_s^{\alpha} = \frac{\eta_s^{\alpha} \bar{\rho}_s}{\bar{\rho}_s^{\alpha}}. \quad (142c)$$

The solid Helmholtz energy is the mass-fraction average (80), while the solid skeleton stress is the sum of the component partial stresses. The intrinsic skeleton stress is obtained by normalizing by the solid volume fraction:

$$\boldsymbol{\sigma}'_s = \sum_{\alpha} \boldsymbol{\sigma}'_s{}^{\alpha} = \phi_s \sum_{\alpha} \phi_s^{\alpha} \bar{\boldsymbol{\sigma}}_s'^{\alpha}, \quad (143a)$$

$$\boldsymbol{\sigma}'_s{}^{\alpha} = \rho_s^{\alpha} \left. \frac{\partial \psi_s^{\alpha}}{\partial \bar{\mathbf{F}}_s^M} \right|_{\mathcal{S}} (\bar{\mathbf{F}}_s^M)^T, \quad (143b)$$

$$\bar{\boldsymbol{\sigma}}_s'^{\alpha} = \bar{\rho}_s^{\alpha} \left. \frac{\partial \psi_s^{\alpha}}{\partial \bar{\mathbf{F}}_s^M} \right|_{\mathcal{S}} (\bar{\mathbf{F}}_s^M)^T, \quad (143c)$$

For the pressure and Biot split below, the Coleman–Noll calculation resolves solid density dependence through the physical intrinsic density $\bar{\rho}_s^{\alpha}$ of each solid component. With this specialization, the intrinsic solid pressure is the component-volume-fraction average of the component pressures,

$$\bar{p}_s = \sum_{\alpha} \phi_s^{\alpha} \bar{p}_s^{\alpha}, \quad (144a)$$

$$\bar{p}_s^{\alpha} = (\bar{\rho}_s^{\alpha})^2 \left. \frac{\partial \psi_s^{\alpha}}{\partial \bar{\rho}_s^{\alpha}} \right|_{\bar{\mathbf{F}}_s^M, J_s, \theta, \eta_s^{\beta}}, \quad (144b)$$

The distension force is the separate Coleman–Noll restriction

$$\frac{1}{3} \text{tr}(\boldsymbol{\sigma}'_s) = \rho_s a_s \sum_{\alpha} \eta_s^{\alpha} \left. \frac{\partial \psi_s^{\alpha}}{\partial a_s} \right|_{\mathcal{S}}. \quad (145)$$

These are not independent closure equations for double-prime stress. They are mixture rules generated by the same Helmholtz energies that supply entropy, temperature dependence, composition dependence, and the reversible distension response.

The same structure gives the overall momentum equation. Summing the phase momentum balances, using the mobile-phase pressure relations, and retaining the conversion-momentum terms gives

$$\sum_{\xi} \rho_{\xi} \mathbf{a}_{\xi} = \nabla_{\mathbf{x}} \cdot \boldsymbol{\sigma}'_s + \rho \mathbf{g} + \sum_{\xi} \mathbf{b}_{\xi} - \phi_s \nabla_{\mathbf{x}} \bar{p}_E - \sum_l \phi_l \nabla_{\mathbf{x}} \bar{p}_l + \sum_{\xi} \sum_{\alpha} \dot{c}_{\xi}^{\alpha} (\nabla_{\mathbf{x}} \mu_{\xi}^{\alpha} - \mathbf{v}_{\xi}), \quad (146)$$

where l ranges over the mobile pore phases and $\rho = \sum_{\xi} \rho_{\xi}$. This is the pressure form of the capillary force. The primitive potentials γ_l are useful in the variation, but the volume-fraction Euler–Lagrange relation gives $\lambda + \bar{p}_l - \gamma_l = 0$ and therefore $\phi_l (\nabla_{\mathbf{x}} \lambda - \nabla_{\mathbf{x}} \gamma_l) = -\phi_l \nabla_{\mathbf{x}} \bar{p}_l$. The final balance is therefore written in terms of phase pressure gradients, which is the form used in compositional flow. For a closed mixture with pairwise internal interactions,

$$\sum_{\xi} \mathbf{b}_{\xi} = \mathbf{0}. \quad (147)$$

The same Euler–Lagrange relations also give the pressure-gradient collapse needed to compare with the equivalent-pressure form used in multiphase poromechanics. Since $\phi \gamma(S_a, S_g)$ is homogeneous of degree one in the primitive mobile volume fractions, Euler’s theorem gives $\phi \gamma = \sum_l \phi_l \gamma_l$, or $\gamma = \sum_l S_l \gamma_l$. Taking the differential of this Euler relation and subtracting the defining differential $d(\phi \gamma) = \sum_l \gamma_l d\phi_l$ gives the Gibbs–Duhem identity $\sum_l \phi_l d\gamma_l = 0$ [1]. In gradient form, $\sum_l S_l \nabla_{\mathbf{x}} \gamma_l = \mathbf{0}$. Therefore, since $\bar{p}_l = \gamma_l + \bar{p}_E$,

$$\sum_l S_l \nabla_{\mathbf{x}} \bar{p}_l = \nabla_{\mathbf{x}} \bar{p}_E. \quad (148)$$

Using this identity in (146) gives

$$\sum_{\xi} \rho_{\xi} \mathbf{a}_{\xi} = \nabla_{\mathbf{x}} \cdot \boldsymbol{\sigma}'_s + \rho \mathbf{g} - \nabla_{\mathbf{x}} \bar{p}_E + \sum_{\xi} \sum_{\alpha} \dot{c}_{\xi}^{\alpha} (\nabla_{\mathbf{x}} \mu_{\xi}^{\alpha} - \mathbf{v}_{\xi}). \quad (149)$$

If external momentum is supplied to the mixture, the omitted term $\sum_{\xi} \mathbf{b}_{\xi}$ is added to the right-hand side.

To connect this result with finite-deformation poromechanics, do the pressure transform at the solid-component level. Unlike a miscible fluid phase, a solid skeleton component is a physical constituent with its own intrinsic density, current component partial density, and corresponding specific volumes,

$$\bar{v}_s^{\alpha} = \frac{1}{\bar{\rho}_s^{\alpha}}, \quad (150a)$$

$$v_s^{\alpha} = \frac{1}{\rho_s^{\alpha}}, \quad (150b)$$

$$v_{s0}^{\alpha} = \frac{1}{\rho_{s0}^{\alpha}}, \quad (150c)$$

$$\bar{v}_{s0}^{\alpha} = \frac{1}{\bar{\rho}_{s0}^{\alpha}}. \quad (150d)$$

The Coleman–Noll density coefficient has already identified the thermodynamic component pressure in (144b). Equivalently, since $\bar{v}_s^\alpha = 1/\bar{\rho}_s^\alpha$,

$$\bar{p}_s^\alpha = - \left. \frac{\partial \psi_s^\alpha}{\partial \bar{v}_s^\alpha} \right|_{\bar{\mathbf{F}}_s^M, J_s, \theta, \eta_s^\beta}. \quad (151)$$

The storage multipliers may be normalized by the current volume occupied by each solid component,

$$\bar{p}_s^\alpha = \frac{\tau_s^\alpha}{\phi_s \phi_s^\alpha}. \quad (152)$$

This is a normalization of the multiplier, not a separate variation with respect to $\bar{\rho}_s^\alpha$; the mixture pressure is still the average in (144a). The component-density dependence has already been accounted for in the Coleman–Noll chain rule. The component Legendre transform is therefore

$$\tilde{\psi}_s^\alpha = \psi_s^\alpha + \bar{p}_s^\alpha v_s^\alpha, \quad (153a)$$

$$\left. \frac{\partial \tilde{\psi}_s^\alpha}{\partial \bar{p}_s^\alpha} \right|_{\bar{\mathbf{F}}_s^M, J_s, \theta, \eta_s^\beta} = v_s^\alpha. \quad (153b)$$

The pressure in the saturation constraint appears only after the component pressures are summed. Since the solid phase carries no mobile capillary potential, (48) gives

$$\phi_s \lambda = - \sum_{\alpha} \tau_s^\alpha, \quad (154a)$$

$$\bar{p}_E \equiv -\lambda = \frac{1}{\phi_s} \sum_{\alpha} \tau_s^\alpha = \sum_{\alpha} \phi_s^\alpha \bar{p}_s^\alpha. \quad (154b)$$

Thus the equivalent pressure is the component-volume-fraction average of the component pressures; it is not the pressure used to perform the component Legendre transforms. The corresponding stress split follows by differentiating the component Legendre transform with respect to the mechanical deformation. The material component mass relation (6) shows that, when the conversion and composition variables are held fixed in this derivative, the density dependence induced by $\bar{\mathbf{F}}_s^M$ enters through the Jacobian. Equivalently, the component specific volume may be treated as $v_s^\alpha = v_s^\alpha(J_s, \bar{p}_s^\alpha, \theta, \eta_s^\beta)$ for this calculation. Therefore

$$\left. \frac{\partial v_s^\alpha}{\partial \bar{\mathbf{F}}_s^M} \right|_{\bar{p}_s^\alpha, \theta, \eta_s^\beta} = \left. \frac{\partial v_s^\alpha}{\partial J_s} \right|_{\bar{p}_s^\alpha, \theta, \eta_s^\beta} \frac{\partial J_s}{\partial \bar{\mathbf{F}}_s^M} \quad (155)$$

$$= \left. \frac{\partial v_s^\alpha}{\partial J_s} \right|_{\bar{p}_s^\alpha, \theta, \eta_s^\beta} J_s (\bar{\mathbf{F}}_s^M)^{-T}. \quad (156)$$

At fixed $\bar{p}_s^\alpha$, the Legendre transform gives

$$\rho_{s0}^\alpha \left. \frac{\partial \psi_s^\alpha}{\partial \bar{\mathbf{F}}_s^M} \right|_S = \rho_{s0}^\alpha \left. \frac{\partial \tilde{\psi}_s^\alpha}{\partial \bar{\mathbf{F}}_s^M} \right|_{\bar{p}_s^\alpha, \theta, \eta_s^\beta} - \bar{p}_s^\alpha \frac{1}{v_{s0}^\alpha} \left. \frac{\partial v_s^\alpha}{\partial \bar{\mathbf{F}}_s^M} \right|_{\bar{p}_s^\alpha, \theta, \eta_s^\beta}. \quad (157)$$

The first two terms define the component first Piola–Kirchhoff stresses,

$$\mathbf{P}_s'^\alpha = \rho_{s0}^\alpha \left. \frac{\partial \psi_s^\alpha}{\partial \bar{\mathbf{F}}_s^M} \right|_S, \quad \mathbf{P}_s''^\alpha = \rho_{s0}^\alpha \left. \frac{\partial \tilde{\psi}_s^\alpha}{\partial \bar{\mathbf{F}}_s^M} \right|_{\bar{p}_s^\alpha, \theta, \eta_s^\beta}. \quad (158)$$

Thus

$$\mathbf{P}'_\alpha = \mathbf{P}''_\alpha - \bar{p}_s^\alpha \frac{1}{v_{s0}^\alpha} \frac{\partial v_s^\alpha}{\partial J_s} \bigg|_{\bar{p}_s^\alpha, \theta, \eta_s^\beta} J_s (\bar{\mathbf{F}}_s^M)^{-T}. \quad (159)$$

Pushing forward with $\boldsymbol{\sigma} = J_s^{-1} \mathbf{P} (\bar{\mathbf{F}}_s^M)^T$ gives

$$\boldsymbol{\sigma}'_\alpha = \boldsymbol{\sigma}''_\alpha - \frac{1}{v_{s0}^\alpha} \frac{\partial v_s^\alpha}{\partial J_s} \bigg|_{\bar{p}_s^\alpha, \theta, \eta_s^\beta} \bar{p}_s^\alpha \mathbf{I}. \quad (160)$$

This identifies the component nonlinear Biot coefficient as

$$B_s^\alpha \equiv 1 - \frac{1}{v_{s0}^\alpha} \frac{\partial v_s^\alpha}{\partial J_s} \bigg|_{\bar{p}_s^\alpha, \theta, \eta_s^\beta}. \quad (161)$$

$$\approx 1 - \frac{K^{\star\alpha}}{K_s^\alpha} \quad (162)$$

With this definition, the transformed component stress split is

$$\boldsymbol{\sigma}'_\alpha = \boldsymbol{\sigma}''_\alpha + (1 - B_s^\alpha) \bar{p}_s^\alpha \mathbf{I}. \quad (163)$$

Substitution into the Coleman–Noll mixture rule gives the solid partial stress split

$$\boldsymbol{\sigma}'_s = \sum_\alpha \boldsymbol{\sigma}''_\alpha + \sum_\alpha (1 - B_s^\alpha) \bar{p}_s^\alpha \mathbf{I}, \quad (164a)$$

$$\boldsymbol{\sigma}'_s \equiv \boldsymbol{\sigma}''_s + (1 - B_s) \bar{p}_E \mathbf{I}, \quad (164b)$$

$$1 - B_s = \frac{1}{\bar{p}_E} \sum_\alpha (1 - B_s^\alpha) \bar{p}_s^\alpha. \quad (164c)$$

Thus the solid mixture Biot coefficient is obtained only after the component pressure contributions have been summed. In this partial-stress form the component stresses already carry the current component partial density ρ_s^α , rather than an additional intrinsic volume-fraction weight.

Combining this equivalent-pressure form with (164) gives

$$\sum_\xi \rho_\xi \mathbf{a}_\xi = \nabla_{\mathbf{x}} \cdot [\boldsymbol{\sigma}''_s + (1 - B_s) \bar{p}_E \mathbf{I}] + \rho \mathbf{g} - \nabla_{\mathbf{x}} \bar{p}_E + \sum_\xi \sum_\alpha \dot{c}_\xi^\alpha (\nabla_{\mathbf{x}} \mu_\xi^\alpha - \mathbf{v}_\xi). \quad (165)$$

Collecting the equivalent-pressure term into the total stress,

$$\sum_\xi \rho_\xi \mathbf{a}_\xi = \nabla_{\mathbf{x}} \cdot [\boldsymbol{\sigma}''_s - B_s \bar{p}_E \mathbf{I}] + \rho \mathbf{g} + \sum_\alpha \sum_\xi \dot{c}_\xi^\alpha (\nabla_{\mathbf{x}} \mu_\xi^\alpha - \mathbf{v}_\xi). \quad (166)$$

When inertia and conversion-momentum production are neglected, this is the quasistatic total-stress balance used by Kim, Tchelepi, and Juanes, with $\bar{p}_E$ replacing the saturation-weighted average pressure.

3.3 Distension stress and crystallization pressure in practice

Crystallization pressure is not introduced here as an additional load in the momentum equation. The mechanical response must first be generated by an admissible solid free energy. The Coleman–Noll restriction (145) states that the hydrostatic part of the total solid stress is the force conjugate to the solid distension:

$$\frac{1}{3} \text{tr}(\boldsymbol{\sigma}'_s) = \rho_s a_s \sum_\alpha \eta_s^\alpha \frac{\partial \psi_s^\alpha}{\partial a_s} \bigg|_{\mathcal{S}}. \quad (167)$$

Using the equivalent-pressure Biot split (164), this same condition can be written as a restriction on the double-prime stress:

$$\frac{1}{3} \text{tr}(\boldsymbol{\sigma}_s'') = \rho_s a_s \sum_{\alpha} \eta_s^{\alpha} \left. \frac{\partial \psi_s^{\alpha}}{\partial a_s} \right|_{\mathcal{S}} - (1 - B_s) \bar{p}_E. \quad (168)$$

Equations (167) and (168) are not extra balance laws for a_s ; they are admissibility conditions on the reversible solid stress. The solid volume ratios satisfy

$$J_s = a_s \bar{J}_s, \quad \ln J_s = \ln a_s + \ln \bar{J}_s, \quad \ln \left(\frac{J_s}{a_s} \right) = \ln \bar{J}_s. \quad (169)$$

Thus a penalty written only in terms of $\ln a_s$ or only in terms of $\ln(J_s/a_s) = \ln \bar{J}_s$ will not, in general, produce the trace required by Coleman–Noll. The compatible elastic pressure scale must be tied to $\ln J_s$, because J_s is the product that changes with both the distension and the deformation Jacobian.

The component Legendre transform was defined in (153a) as $\tilde{\psi}_s^{\alpha} = \psi_s^{\alpha} + \bar{p}_s^{\alpha} v_s^{\alpha}$. Thus the transformed potential generates the double-prime stress, while the untransformed Helmholtz energy generates the total stress. The difference between the two stress measures is exactly the Biot pressure correction in (163). Therefore the Biot pressure contribution should not be inserted a second time into the transformed volumetric energy. For a solid component α , take the current-volume transformed volumetric stress potential to be

$$\tilde{W}_{\text{vol}}^{\alpha} = \frac{K^{\star\alpha}}{2} (\ln J_s)^2. \quad (170)$$

The admissibility check is direct. Holding $\bar{J}_s$ and $\bar{p}_s^{\alpha}$ fixed,

$$\left. \frac{\partial \tilde{W}_{\text{vol}}^{\alpha}}{\partial a_s} \right|_{\bar{J}_s, \bar{p}_s^{\alpha}} = \frac{K^{\star\alpha}}{a_s} \ln J_s. \quad (171)$$

Therefore

$$a_s \left. \frac{\partial \tilde{W}_{\text{vol}}^{\alpha}}{\partial a_s} \right|_{\bar{J}_s, \bar{p}_s^{\alpha}} = K^{\star\alpha} \ln J_s. \quad (172)$$

This is exactly the hydrostatic double-prime stress generated by the transformed volumetric potential:

$$\frac{1}{3} \text{tr}(\boldsymbol{\sigma}_s''^{\alpha}) = K^{\star\alpha} \ln J_s. \quad (173)$$

To return to the original Helmholtz derivative that appears in the Coleman–Noll restriction, use the same Legendre/Biot relation in scalar trace form:

$$\rho_s a_s \eta_s^{\alpha} \left. \frac{\partial \psi_s^{\alpha}}{\partial a_s} \right|_{\mathcal{S}} = a_s \left. \frac{\partial \tilde{W}_{\text{vol}}^{\alpha}}{\partial a_s} \right|_{\bar{J}_s, \bar{p}_s^{\alpha}} + (1 - B_s^{\alpha}) \bar{p}_s^{\alpha}. \quad (174)$$

Substituting (172) into (174) gives the component total trace:

$$\frac{1}{3} \text{tr}(\boldsymbol{\sigma}_s'^{\alpha}) = K^{\star\alpha} \ln J_s + (1 - B_s^{\alpha}) \bar{p}_s^{\alpha}. \quad (175)$$

Equivalently, the component Coleman–Noll distension force is

$$\rho_s a_s \eta_s^{\alpha} \left. \frac{\partial \psi_s^{\alpha}}{\partial a_s} \right|_{\mathcal{S}} = K^{\star\alpha} \ln J_s + (1 - B_s^{\alpha}) \bar{p}_s^{\alpha}. \quad (176)$$

Summing over solid components gives the total trace condition:

$$\frac{1}{3} \text{tr}(\boldsymbol{\sigma}'_s) = \rho_s a_s \sum_{\alpha} \eta_s^{\alpha} \left. \frac{\partial \psi_s^{\alpha}}{\partial a_s} \right|_{\mathcal{S}}, \quad (177)$$

which is (167). Subtracting the Biot equivalent-pressure contribution recovers the double-prime condition:

$$\frac{1}{3} \text{tr}(\boldsymbol{\sigma}''_s) = \sum_{\alpha} K^{\star\alpha} \ln J_s = \rho_s a_s \sum_{\alpha} \eta_s^{\alpha} \left. \frac{\partial \psi_s^{\alpha}}{\partial a_s} \right|_{\mathcal{S}} - (1 - B_s) \bar{p}_E, \quad (178)$$

Thus (170) conforms to the Coleman–Noll admissibility condition: the same transformed volumetric potential that supplies the double-prime stress also supplies the distension force required by (167). Once this stress is used in the momentum equation, no separate crystallization-pressure force should be added.

It is still useful to name the pressure scale carried by the transformed volumetric energy:

$$p_{\text{xtal}}^{\alpha} \equiv K^{\star\alpha} \ln J_s, \quad (179)$$

Then

$$\frac{1}{3} \text{tr}(\boldsymbol{\sigma}'_s{}^{\alpha}) = p_{\text{xtal}}^{\alpha} + (1 - B_s^{\alpha}) \bar{p}_s^{\alpha}. \quad (180)$$

This mechanical pressure scale is not the kinetic driving force. Crystallization enters the thermodynamics through the mechanism affinity. The variational theory identifies the force conjugate to the mechanism progress rate $\dot{r}_m$:

$$\delta W_{\text{conv}} = - \sum_m \int_{\mathcal{B}} \mathcal{A}_m \delta \dot{r}_m \, dv, \quad \mathcal{A}_m = - \sum_{\xi} \sum_{\alpha} \mu_{\xi}^{\alpha} \nu_{\xi m}^{\alpha}. \quad (181)$$

The entropy inequality requires the kinetic closure to make this solved driving force admissible:

$$\mathcal{A}_m \dot{r}_m \geq 0. \quad (182)$$

A simple admissible choice is $\dot{r}_m = M_m \mathcal{A}_m$ with $M_m \geq 0$. A one-sided precipitation or dissolution law may be used when the mechanism is not reversible.

Classical crystallization pressure is recovered only after this thermodynamic solve. For a mechanism m precipitating solid component α_m , define the intrinsic specific volume

$$\bar{v}_s^{\alpha_m} = \frac{1}{\bar{\rho}_s^{\alpha_m}}. \quad (183)$$

The equivalent crystallization pressure is the pressure scale obtained by combining the chemical affinity with the admissible mechanical trace stress:

$$p_c^{(m)} = \frac{\mathcal{A}_m}{\bar{v}_s^{\alpha_m}} + \frac{1}{3} \text{tr}(\boldsymbol{\sigma}'_s{}^{\alpha_m}). \quad (184)$$

Equivalently,

$$\mathcal{A}_m = \bar{v}_s^{\alpha_m} \left(p_c^{(m)} - \frac{1}{3} \text{tr}(\boldsymbol{\sigma}'_s{}^{\alpha_m}) \right). \quad (185)$$

Using (180), this diagnostic pressure is

$$p_c^{(m)} = \frac{\mathcal{A}_m}{\bar{v}_s^{\alpha_m}} + p_{\text{xtal}}^{\alpha_m} + (1 - B_s^{\alpha_m}) \bar{p}_s^{\alpha_m}. \quad (186)$$

At reversible equilibrium,

$$p_c^{(m)} = p_{\text{xtal}}^{\alpha_m} + (1 - B_s^{\alpha_m}) \bar{p}_s^{\alpha_m}. \quad (187)$$

Thus crystallization pressure arises as a diagnostic pressure scale obtained from the solved affinity and the admissible mechanical trace stress. It is not an independent constitutive stress. A classical supersaturation law can be used by identifying

$$p_c^{(m)} = \frac{R\theta}{V_s^{\alpha_m}} \ln \Omega_m, \quad (188)$$

where $V_s^{\alpha_m}$ is the molar volume of the precipitating solid component. The saturation ratio associated with the solved thermodynamic state is then

$$\Omega_m = \exp \left(\frac{V_s^{\alpha_m}}{R\theta} \left[\frac{\mathcal{A}_m}{\bar{v}_s^{\alpha_m}} + \frac{1}{3} \text{tr}(\boldsymbol{\sigma}'^{\alpha_m}) \right] \right). \quad (189)$$

In this form, a classical crystallization-pressure formula supplies either a diagnostic output or a calibration target for the kinetic law. The simulator still advances the Helmholtz-generated stresses, balance laws, conversion constraints, and an admissible law for $\dot{r}_m$; it does not add $p_c^{(m)}$ as a second mechanical load.

3.4 Model summary, closures, and equation count

This section collects the equations that would be assembled in a simulator. The point of the summary is to separate primary unknowns, reversible constitutive relations generated by the Helmholtz energies, and genuinely dissipative closure laws. The component mass balances are

$$\frac{\partial}{\partial t} (\phi_\xi \bar{\rho}_\xi \eta_\xi^\alpha) + \nabla_{\mathbf{x}} \cdot (\phi_\xi \bar{\rho}_\xi \eta_\xi^\alpha \mathbf{v}_\xi) = \sum_m \nu_{\xi m}^\alpha \dot{r}_m. \quad (190)$$

The phase momentum balances may be kept phasewise in the form (49), with the mobile-phase pressure-gradient specialization shown in (67)–(69), or summed into the overall poromechanics form (166). The algebraic volume and composition constraints are

$$\sum_\xi \phi_\xi = 1, \quad (191a)$$

$$\sum_\alpha \eta_\xi^\alpha = 1. \quad (191b)$$

with the corresponding Euler–Lagrange relations

$$\lambda + \bar{p}_l - \gamma_l = 0, \quad l \text{ mobile}, \quad (192a)$$

$$\lambda + \bar{p}_s = 0, \quad (192b)$$

$$\sum_\alpha \frac{\tau_\xi^\alpha}{\phi_\xi} = \bar{p}_\xi. \quad (192c)$$

The composition stationarity equations are used in their projected form, with the redundant normal multiplier π_ξ eliminated explicitly. For mobile phases this is (115); for the solid it is (116), where the component-density chain-rule and stress-free compositional-dilation corrections are retained. The intrinsic pressures, solid stresses, entropy, and distension stress are not independent laws; they are Coleman–Noll derivatives of the Helmholtz energies summarized by (117), (121), (127), and (128).

The conversion potentials are solved from (64). After the entropy argument identifies the uniform source-carried closure $L_\xi^\alpha = \mu_\xi^\alpha - \hat{\psi}_\xi^\alpha$, with $\hat{\psi}_l^\alpha = \psi_l$ for mobile phases and $\hat{\psi}_s^\alpha = \psi_s^\alpha$ for the component-additive solid, the mechanism affinity is the stoichiometric projection

$$\mathcal{A}_m = - \sum_\xi \sum_\alpha \mu_\xi^\alpha \nu_{\xi m}^\alpha. \quad (193)$$

The remaining closure laws are dissipative. They must be chosen so that

$$\sum_m \mathcal{A}_m \dot{r}_m + \sum_{\xi < \zeta} \mathbf{u}_{\xi\zeta} \cdot \mathbf{k}_{\xi\zeta}^{-1} \mathbf{u}_{\xi\zeta} + \sum_\xi \frac{\nabla_{\mathbf{x}} \theta \cdot \mathbf{K}_\xi \nabla_{\mathbf{x}} \theta}{\theta} \geq 0. \quad (194)$$

For example, one may use kinetic laws $\dot{r}_m = \mathcal{K}_m(\mathcal{A}_m, \text{state})$ with $\mathcal{A}_m \dot{r}_m \geq 0$, pairwise interaction laws (138), and the heat-flux closure (140). A crystallization-pressure law, if used, enters only by translating it into the affinity through (185).

Let P be the number of phases, N the number of components tracked in each phase, N_m the number of independent conversion mechanisms, and d the spatial dimension. Using phase displacements as the kinematic unknowns, with velocities obtained by time differentiation, a convenient fully implicit pointwise unknown accounting is

Unknown family	Count
Phase displacements $u_{\xi 1}, \dots, u_{\xi d}$ for $\xi = 1, \dots, P$	dP
Intrinsic phase densities $\bar{\rho}_\xi$ for $\xi = 1, \dots, P$	P
Phase mass fractions $\eta_\xi^1, \dots, \eta_\xi^N$ for $\xi = 1, \dots, P$	PN
Phase volume fractions ϕ_ξ for $\xi = 1, \dots, P$	P
Saturation multiplier λ	1
Storage multipliers $\tau_\xi^1, \dots, \tau_\xi^N$ for $\xi = 1, \dots, P$	PN
Helmholtz chemical potentials $\mu_\xi^1, \dots, \mu_\xi^N$ for $\xi = 1, \dots, P$	PN
Mechanism rates $\dot{r}_1, \dots, \dot{r}_{N_m}$	N_m
Temperature θ	1

The corresponding scalar unknown count is

$$N_{\text{unk}} = dP + P + PN + P + 1 + PN + PN + N_m + 1 = dP + 3PN + 2P + N_m + 2. \quad (195)$$

The same number of scalar equations is supplied by the following blocks:

Equation block	Count
Component mass balances (190)	PN
Phase momentum balances (49)	dP
Saturation constraint in (191)	1
Volume-fraction Euler–Lagrange relations in (192)	P
Intrinsic-density relations in (192)	P
Projected composition stationarity (115) and (116)	$P(N - 1)$
Composition normalizations in (191)	P
Conversion-potential equations (64)	PN
Kinetic laws for $\dot{r}_m$	N_m
Energy or temperature equation using (104)	1

Thus

$$N_{\text{eq}} = PN + dP + 1 + P + P + P(N - 1) + P + PN + N_m + 1 = dP + 3PN + 2P + N_m + 2 = N_{\text{unk}}. \quad (196)$$

This count treats π_ξ , $\mathcal{A}_m$, and $\dot{c}_\xi^\alpha$ as eliminated quantities: π_ξ is removed by projecting the composition stationarity relations onto the $N - 1$ dimensional tangent space of $\sum_\alpha \eta_\xi^\alpha = 1$, $\mathcal{A}_m$ is substituted from (193), and $\dot{c}_\xi^\alpha$ is substituted from the stoichiometric source term in (190). If any of these are retained as auxiliary unknowns, the same number of defining equations is added, so the count is unchanged. For an isothermal model, remove θ and the energy equation from both sides of the count.

3.5 Relative flux vectors

The phase momentum equations above determine consistent Darcy-scale relative fluxes once the interphase drag is written in terms of a velocity relative to the solid skeleton. For each mobile phase l , define the current relative mass flux by

$$\mathbf{w}_l = \phi_l \bar{\rho}_l (\mathbf{v}_l - \mathbf{v}_s) = \rho_l (\mathbf{v}_l - \mathbf{v}_s). \quad (197)$$

Thus

$$\mathbf{v}_l = \mathbf{v}_s + \frac{\mathbf{w}_l}{\rho_l}. \quad (198)$$

The same linear-drag argument used in the reacting-mixture formulation closes the part of the mobile-phase interaction force produced by motion through the skeleton as

$$\mathbf{b}_l = -\frac{\mu_l \phi_l}{\bar{\rho}_l} \boldsymbol{\kappa}_l^{-1} \mathbf{w}_l, \quad (199)$$

where μ_l is the dynamic viscosity and $\boldsymbol{\kappa}_l$ is the permeability tensor for phase l . Phase relative permeability may be absorbed into $\boldsymbol{\kappa}_l$.

Substituting (198) and (199) into the mobile momentum balance (67)–(69) gives, for each mobile phase,

$$\begin{aligned} \frac{1}{\rho_l} \left(\phi_l^2 \mu_l \boldsymbol{\kappa}_l^{-1} + \sum_\alpha \dot{c}_l^\alpha \mathbf{I} \right) \mathbf{w}_l &= \rho_l (\mathbf{g} - \mathbf{a}_l) - \phi_l \nabla_{\mathbf{x}} \bar{p}_l \\ &+ \sum_\alpha \dot{c}_l^\alpha (\nabla_{\mathbf{x}} \mu_l^\alpha - \mathbf{v}_s), \end{aligned} \quad (200)$$

The term $\sum_\alpha \dot{c}_l^\alpha \mathbf{I}$ is the same reaction-induced correction that appears in the three-phase reacting-mixture flux law. It comes from the velocity carried inside the conversion-momentum term, not from appending a source to the mass balance after the fact.

Define the modified permeability tensor

$$\tilde{\boldsymbol{\kappa}}_l = \left(\phi_l^2 \mu_l \boldsymbol{\kappa}_l^{-1} + \sum_\alpha \dot{c}_l^\alpha \mathbf{I} \right)^{-1}. \quad (201)$$

Then the admissible relative mass flux is

$$\mathbf{w}_l = \rho_l \tilde{\boldsymbol{\kappa}}_l \left[\rho_l (\mathbf{g} - \mathbf{a}_l) - \phi_l \nabla_{\mathbf{x}} \bar{p}_l + \sum_\alpha \dot{c}_l^\alpha (\nabla_{\mathbf{x}} \mu_l^\alpha - \mathbf{v}_s) \right]. \quad (202)$$

This expression is the multicomponent mobile-phase analogue of the generalized Darcy flux in the earlier reacting-mixture model. The first two terms are the usual body-force and pressure-gradient contributions. The last term is the thermodynamic conversion correction: mass transfer into or

out of phase l drives relative motion through gradients of the solved component potentials μ_l^α and through the momentum carried by the solid frame.

When conversion is absent, $\dot{c}_l^\alpha = 0$, and when mobile-phase inertia is negligible, (202) reduces to

$$\phi_l (\mathbf{v}_l - \mathbf{v}_s) = -\frac{\kappa_l}{\mu_l} (\nabla_{\mathbf{x}} \bar{p}_l - \bar{\rho}_l \mathbf{g}). \quad (203)$$

which is Darcy's law for phase l relative to the deforming solid skeleton. In the reacting case, (202) should be used instead of (203) whenever conversion rates are large enough for momentum production or gradients of μ_l^α to affect transport.

3.6 Classical compositional-flow limit and interpretation of the multipliers

The preceding equations contain the usual compositional-reservoir structure as a special limit. The phase/component source is written as

$$\dot{c}_\xi^\alpha = \sum_m \nu_{\xi m}^\alpha \dot{r}_m, \quad (204)$$

which is the phase-resolved analogue of the standard reactive-transport source vector $\mathbf{R} = \nu \mathbf{q}$. Summing (13) over phases gives

$$\frac{\partial}{\partial t} \left(\sum_\xi \phi_\xi \bar{\rho}_\xi \eta_\xi^\alpha \right) + \nabla_{\mathbf{x}} \cdot \left(\sum_\xi \phi_\xi \bar{\rho}_\xi \eta_\xi^\alpha \mathbf{v}_\xi \right) = \sum_\xi \sum_m \nu_{\xi m}^\alpha \dot{r}_m. \quad (205)$$

To see the usual reservoir form, restrict attention to the mobile pore phases and write their volume fractions as

$$\phi_\xi = \phi S_\xi, \quad \sum_{\xi \in \{a, o, g\}} S_\xi = 1, \quad (206)$$

where ϕ is porosity and S_ξ is the saturation of phase ξ . Let M^α be the molar mass of component α , and define the phase component molar concentration per phase volume by

$$\bar{C}_\xi^\alpha = \frac{\bar{\rho}_\xi \eta_\xi^\alpha}{M^\alpha}. \quad (207)$$

Dividing (13) by M^α gives the phase-resolved molar balance

$$\frac{\partial}{\partial t} \left(\phi S_\xi \bar{C}_\xi^\alpha \right) + \nabla_{\mathbf{x}} \cdot \left(\phi S_\xi \bar{C}_\xi^\alpha \mathbf{v}_\xi \right) = \sum_m \frac{\nu_{\xi m}^\alpha}{M^\alpha} \dot{r}_m. \quad (208)$$

If the porous skeleton is fixed, or if the equation is written relative to the skeleton and the advective solid term is moved to the left-hand storage operator, the phase Darcy flux is

$$\mathbf{q}_\xi = \phi S_\xi (\mathbf{v}_\xi - \mathbf{v}_s). \quad (209)$$

For the fixed-skeleton case $\mathbf{v}_s = \mathbf{0}$, so (208) becomes

$$\frac{\partial}{\partial t} \left(\phi S_\xi \bar{C}_\xi^\alpha \right) + \nabla_{\mathbf{x}} \cdot \left(\bar{C}_\xi^\alpha \mathbf{q}_\xi \right) = \sum_m \frac{\nu_{\xi m}^\alpha}{M^\alpha} \dot{r}_m. \quad (210)$$

This is the phase/component conservation equation used before the phase equations are summed in a compositional simulator. A standard Darcy closure supplies

$$\mathbf{q}_\xi = -\frac{k k_{r\xi}}{\mu_\xi} (\nabla_{\mathbf{x}} \bar{p}_\xi - \bar{\rho}_\xi \mathbf{g}), \quad (211)$$

possibly with capillary-pressure relations between the phase pressures. In the present theory this is not postulated at the balance-law level: it is a closure that replaces the mobile-phase momentum equations (67)–(69) after inertia, body interactions, and conversion-momentum terms have been specialized.

Summing (210) over mobile phases yields

$$\frac{\partial}{\partial t} \left(\phi \sum_{\xi} S_{\xi} \bar{C}_{\xi}^{\alpha} \right) + \nabla_{\mathbf{x}} \cdot \left(\sum_{\xi} \bar{C}_{\xi}^{\alpha} \mathbf{q}_{\xi} \right) = \sum_{\xi} \sum_m \frac{\nu_{\xi m}^{\alpha}}{M^{\alpha}} \dot{r}_m. \quad (212)$$

This is the conservative component balance used in compositional reservoir simulation. The usual simulator source for component α is therefore the phase-summed stoichiometric projection on the right-hand side of (212). The difference is that the present coefficients retain the phase label. That phase label permits a single mechanism to encode both chemical conversion and the phase in which the product appears. A pure phase transformation is the special case in which the component label is unchanged and the two nonzero stoichiometric entries are equal and opposite; after summing over phases, its contribution to the total component source vanishes, but it still changes the phase saturations and phase compositions.

The remaining algebraic structure of a traditional compositional simulator is also recovered as a closure of the present variables. The saturation constraint (206) is the mobile-phase version of $\sum_{\xi} \phi_{\xi} = 1$ after the solid volume fraction has been separated. The within-phase composition constraints are

$$\sum_{\alpha} \eta_{\xi}^{\alpha} = 1 \quad \text{or, equivalently,} \quad x_{\xi}^{\alpha} = \frac{\bar{C}_{\xi}^{\alpha}}{\sum_{\beta} \bar{C}_{\xi}^{\beta}}, \quad \sum_{\alpha} x_{\xi}^{\alpha} = 1, \quad (213)$$

This comparison highlights the central thermodynamic distinction. A traditional compositional simulator typically supplies an equation of state, phase equilibrium rules, capillary-pressure relations, component partitioning, and mechanical stress laws as separate closure ingredients. In the present formulation the reversible thermodynamic part is instead generated from the Helmholtz free-energy structure:

$$\psi_{\xi} \implies \bar{p}_{\xi}, \quad s_{\xi}, \quad \boldsymbol{\sigma}'_s, \quad \rho_{\xi} \left. \frac{\partial \psi_{\xi}}{\partial \eta_{\xi}^{\alpha}} \right|_{\dots}, \quad \frac{\tau_{\xi}^{\alpha}}{\eta_{\xi}^{\alpha}}. \quad (214)$$

Thus pressure, entropy, stress, composition derivatives, and normalized component storage potentials are not independent reversible closures; they are conjugate derivatives or multipliers associated with the constrained Helmholtz structure. Transport and reaction kinetics still require dissipative closures, such as mobilities, permeabilities, diffusivities, and reaction-rate laws, but the thermodynamic forces driving those laws are generated by the same free-energy potential.

where x_{ξ}^{α} is the phase mole fraction. Flash calculations then impose phase-equilibrium closures, commonly written as fugacity equalities. In the present notation the reversible version of such a phase transfer for component α from phase ξ to phase ζ has $\nu_{\xi m}^{\alpha} = -1$ and $\nu_{\zeta m}^{\alpha} = +1$, so (220) reduces to

$$\mu_{\zeta}^{\alpha} - \mu_{\xi}^{\alpha} = 0, \quad (215)$$

which is the chemical-potential form of the fugacity-equality condition used in isothermal phase equilibrium.

The multiplier μ_{ξ}^{α} in (21) is the chemical-potential-like variable conjugate to phase/component conversion. Its evolution equation is

$$\frac{D_{\xi} \mu_{\xi}^{\alpha}}{Dt} = \frac{1}{2} \mathbf{v}_{\xi} \cdot \mathbf{v}_{\xi} - \psi_{\xi} - L_{\xi}^{\alpha} - \frac{\tau_{\xi}^{\alpha}}{\phi_{\xi} \bar{\rho}_{\xi} \eta_{\xi}^{\alpha}}. \quad (216)$$

The first term is the kinetic contribution associated with adding mass to a phase moving with velocity $\mathbf{v}_\xi$. The next two terms are the phase Helmholtz free energy and component conversion-work contribution. The last term is the contribution of the component mass constraint; τ_ξ^α is the multiplier enforcing the storage relation between the reference amount of component α and the current partial mass density $\phi_\xi \bar{\rho}_\xi \eta_\xi^\alpha$. After the entropy inequality identifies $L_\xi^\alpha = \mu_\xi^\alpha - \hat{\psi}_\xi^\alpha$, where $\hat{\psi}_l^\alpha = \psi_l$ for mobile phases and $\hat{\psi}_s^\alpha = \psi_s^\alpha$ for the component-additive solid, the closed conversion-potential equations can be written first as

$$\frac{D_\xi \mu_\xi^\alpha}{Dt} = \frac{1}{2} \mathbf{v}_\xi \cdot \mathbf{v}_\xi - \psi_\xi + \hat{\psi}_\xi^\alpha - \mu_\xi^\alpha - \frac{\tau_\xi^\alpha}{\phi_\xi \bar{\rho}_\xi \eta_\xi^\alpha}. \quad (217)$$

Equivalently, separating mobile and solid phases gives

$$\frac{D_l \mu_l^\alpha}{Dt} = \frac{1}{2} \mathbf{v}_l \cdot \mathbf{v}_l - \mu_l^\alpha - \frac{\tau_l^\alpha}{\phi_l \bar{\rho}_l \eta_l^\alpha}, \quad l \text{ mobile}, \quad (218a)$$

$$\frac{D_s \mu_s^\alpha}{Dt} = \frac{1}{2} \mathbf{v}_s \cdot \mathbf{v}_s - \psi_s + \psi_s^\alpha - \mu_s^\alpha - \frac{\tau_s^\alpha}{\phi_s \bar{\rho}_s \eta_s^\alpha}. \quad (218b)$$

Using the component-additive solid rule, the solid Helmholtz difference is $-\psi_s + \psi_s^\alpha = -\sum_\beta \eta_s^\beta \psi_s^\beta + \psi_s^\alpha$. Thus $D_\xi \mu_\xi^\alpha / Dt$ is a generalized chemical-mechanical potential. In a quasi-static isothermal limit with negligible kinetic terms and a conventional choice of thermodynamic potential, this quantity reduces to the chemical potential (or, equivalently in reservoir practice, to a fugacity/equilibrium potential) that appears in phase and reaction equilibrium conditions.

The affinity is the stoichiometric projection of these phase/component potentials:

$$\mathcal{A}_m = - \sum_\xi \sum_\alpha \mu_\xi^\alpha \nu_{\xi m}^\alpha. \quad (219)$$

For an equilibrated mechanism, imposing $\mathcal{A}_m = 0$ gives

$$\sum_\xi \sum_\alpha \mu_\xi^\alpha \nu_{\xi m}^\alpha = 0, \quad (220)$$

which is the classical statement that the stoichiometric sum of chemical potentials vanishes. For a kinetically limited mechanism, such as a slow fluid–solid reaction, (219) instead supplies the driving force in a closure

$$\dot{r}_m = \mathcal{K}_m(\mathcal{A}_m, \text{state}), \quad \mathcal{A}_m \dot{r}_m \geq 0 \quad (221)$$

with the sign convention used here. Thus equilibrium chemistry, kinetic reactive transport, and phase transformation all use the same stoichiometric projection; only the closure for $\dot{r}_m$ changes.

The momentum equations contain an additional conversion term absent from purely kinematic Darcy-scale compositional balances,

$$\sum_\alpha \sum_\xi \dot{c}_\xi^\alpha \left(\nabla_{\mathbf{x}} \mu_\xi^\alpha - \mathbf{v}_\xi \right). \quad (222)$$

The term combines the inertial bookkeeping required when mass is removed from or added to a moving phase with the gradient force generated by variation of the current-rate constraint $\dot{c}_\xi^\alpha = \dot{\mathcal{C}}_\xi^\alpha / J_\xi$. Total mass conservation constrains the component source rates, but the explicit form above avoids introducing an additional conversion-velocity notation. These terms vanish or are hidden in classical reservoir models when phase momenta are not solved explicitly and the phase velocities are supplied by Darcy laws.

Finally, the remaining multipliers have standard constraint interpretations: λ enforces saturation of the pore volume by the phases, π_ξ enforces $\sum_\alpha \eta_\xi^\alpha = 1$, and τ_ξ^α enforces the phase/component storage relation. The novelty of the present formulation is not the stoichiometric source form itself, but the fact that it is embedded in a phase-attached variational mechanics in which the Jacobian J_ξ is varied. That is what exposes the conversion-induced momentum production terms and gives a direct bridge between mixture mechanics and classical compositional reactive transport.

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